Computation of the exact density function of the product of a Wishart matrix and a normal vector

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ABSTRACT

This paper derives the density function of the product of a Wishart matrix and a normal vector that are independently distributed. Unlike existing results, we allow the covariance matrix of the Wishart distribution and that of the normal vector to differ, and we do not assume them to be proportional. In this general setting, the density is shown to admit a two-dimensional integral representation. We further obtain the density of linear combinations of the product. For a single linear combination, it is again expressed as a two-dimensional integral whose integrand involves the modified Bessel function of the second kind, and the special case in which the covariance matrix of the normal vector is proportional to the inverse of the covariance matrix of the Wishart distribution is treated explicitly. In addition, we show that the characteristic function of any number p of linear combinations is a one-dimensional integral, from which the density of $p < m$ linear combinations is recovered as a $(p+1)$-dimensional integral; together with the two-dimensional representation of the full product, this makes the density of any p linear combinations available from an integral of dimension at most $\min(p+1, m-p+2)$, where m is the dimension of the product. All resulting expressions can be evaluated numerically, and their accuracy is illustrated by a comparison with simulation; the product defines a new family of matrix-variate mean–variance mixtures of multivariate normal distributions.

1. Introduction

Both the multivariate normal distribution and the Wishart distribution are fundamental in statistics and are widely used when statistical methods are applied to real data. Given a sample of independent observations from a multivariate normal distribution, the sample mean vector is multivariate normally distributed, the sample covariance matrix follows a Wishart distribution, and the two estimators are independent. These properties have a variety of uses in practice: the independence of the sample mean and covariance matrix is itself a characterizing property of the multivariate normal distribution (see, e.g., Lukacs, 1942), and it is the starting point for many theoretical results, for example in portfolio theory (see, e.g., Jobson and Korkie, 1980; Kan and Zhou, 2007; Kan and Smith, 2008; Bodnar and Schmid, 2009; Bodnar et al., 2016).

In many applications the Wishart and normal distributions appear jointly. The discriminant function (Muirhead, 1982; Bodnar et al., 2020) and the estimated mean-variance portfolio weights (Kan et al., 2022; Bodnar et al., 2023), for example, involve the product of an inverse Wishart matrix and a normal vector (Bodnar and Okhrin, 2011; Bodnar et al., 2022; Kan et al., 2024). The product of a (non-inverse) Wishart matrix and a normal vector, which is the object of this paper, arises just as naturally – for instance when a sample covariance matrix acts on an independent normal vector, or in Bayesian analyses in which the precision matrix and the mean vector carry independent Wishart and normal (semi-conjugate) prior or posterior distributions (see, e.g., Bernardo and Smith, 1994; Hoff, 2009) – and its exact and approximate densities were studied by Bodnar et al. (2013), Bodnar et al. (2015), and Yonenaga and Suzukawa (2023). Recently, Yonenaga (2026) extended these results to the product of a Wishart matrix and an elliptically distributed random vector.

From a probabilistic viewpoint, the product of a Wishart matrix and a normal vector can be regarded as a new family of matrix-variate mixtures of normal distributions, obtained by replacing the univariate generating variable of a classical normal mixture with a random matrix. Normal mixtures based on a univariate generating variable have

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been studied extensively and underlie several well-known families of multivariate distributions, including elliptically contoured distributions (Gupta et al., 2013), skew-normal distributions (Azzalini, 2005, 2013), and the Laplace distribution (Kotz et al., 2012; Kozubowski et al., 2023, 2025). By contrast, the distributional properties of matrix-variate mixtures have received far less attention; some families of matrix-variate elliptically contoured distributions based on such mixtures are discussed in Fang and Zhang (1990).

To make the distinction precise, write $\mathbf{q} = \mathbf{W}\mathbf{z} \overset{d}{=} \mathbf{W}\boldsymbol{\mu} + \mathbf{W}\boldsymbol{\Sigma}_z^{\frac{1}{2}}\boldsymbol{\varepsilon}$, where $\boldsymbol{\varepsilon} \sim N_m(\mathbf{0}_m, \mathbf{I}_m)$ is independent of $\mathbf{W}$, which is Wishart distributed. The symbol $\overset{d}{=}$ denotes the equality in distribution. Conditionally on $\mathbf{W}$, the product is normally distributed with mean vector $\mathbf{W}\boldsymbol{\mu}$ and covariance matrix $\mathbf{W}\boldsymbol{\Sigma}_z\mathbf{W}$: a single random matrix drives both the location, linearly, and the scale, quadratically. This structure differs from that of the classical mixture families in three respects. In the scale mixtures $\boldsymbol{\mu} + \sqrt{R}\,\boldsymbol{\Sigma}^{\frac{1}{2}}\boldsymbol{\varepsilon}$ that generate the elliptically contoured family, and in the mean–variance mixtures $\boldsymbol{\mu} + R\boldsymbol{\gamma} + \sqrt{R}\,\boldsymbol{\Sigma}^{\frac{1}{2}}\boldsymbol{\varepsilon}$ that generate the generalized hyperbolic family (see, e.g., Barndorff-Nielsen et al., 1982), the mixing variable R is a scalar, so that all elements of the random vector share one random scale. In the matrix-variate scale mixtures $\boldsymbol{\mu} + \mathbf{A}\boldsymbol{\varepsilon}$ of Fang and Zhang (1990), the mixing matrix $\mathbf{A}$ is random but the location is fixed. The product $\mathbf{W}\mathbf{z}$ is thus a matrix-variate mean–variance mixture of normal distributions, in which a Wishart matrix takes the role played by the scalar mixing variable in the generalized hyperbolic family. Even in the univariate case $m = 1$ it does not reduce to a generalized hyperbolic law: with $w = \sigma_w^2 v$ and $v \sim \chi_n^2$, we have $q = \sigma_w^2 v(\mu + \sigma_z\varepsilon)$, whose conditional standard deviation, rather than conditional variance, is proportional to the mixing variable.

This paper contributes to the statistical literature in three directions. First, we derive an analytical expression for the density function of the product of a Wishart matrix and a normal vector, extending Bodnar et al. (2013) and Yonenaga and Suzukawa (2023), who provide exact and approximate densities for linear combinations of the product. In particular, we treat the general case in which the covariance matrices of the Wishart and multivariate normal distributions need not be proportional. This substantially extends the framework of Bodnar et al. (2013), where proportional covariance matrices are assumed, while encompassing the more general covariance structure studied by Yonenaga and Suzukawa (2023). The resulting density is expressed as a two-dimensional integral that is easy to implement in practice. The developed approach allows us to derive the joint density of all elements of the product, while the findings of Bodnar et al. (2013) and Yonenaga and Suzukawa (2023) are restricted to linear combinations of the product whose number is strictly smaller than the product dimension. Moreover, for the linear combinations our representations are one dimension lower than those of Bodnar et al. (2013) and Yonenaga and Suzukawa (2023), because the integral over the χ^2-distributed component is evaluated in closed form and yields a modified Bessel function of the second kind; the resulting expressions are therefore easier to evaluate numerically. More importantly, neither of these works delivers the joint density of the full product of a Wishart matrix and a normal vector, which corresponds to the case when the number of linear combinations is equal to the number of elements in the product and is the main object of the present paper. Second, we show that the characteristic function of any number $p \leq m$ of linear combinations of the product is a one-dimensional integral, and that the density of the linear combinations can be recovered from it as a $(p+1)$-dimensional integral which is valid for arbitrary covariance matrices and at every point, including the origin, for every $p < m$, and for $p = m$ when $n > m+1$. Combined with the two-dimensional representation of the full product, the density of any p linear combinations is therefore available as an integral of dimension at most $\min(p+1, m-p+2)$, and of dimension at most three for every p when $\boldsymbol{\Sigma}_z$ is proportional to $\boldsymbol{\Sigma}_w^{-1}$. We also bound the rate at which the characteristic function decays, which yields a simple sufficient condition for the density to be bounded. Third, we introduce a new class of multivariate distributions based on a matrix-variate mean–variance mixture of multivariate normal distributions, extending the existing families of non-normal distributions.

The rest of the paper is organized as follows. The main results are presented in Section 2, where we derive the density of the product of a Wishart matrix and a normal vector. The density of linear combinations of the product, together with several important special cases, is given in Section 3. Section 4 presents the characteristic function approach, and Section 5 validates the representations against simulation and compares their computational cost. Concluding remarks are collected in Section 6, and some technical lemmas are deferred to Appendix A.

2. Distribution of the Product of the Wishart matrix and normally distributed random vector

Suppose $\mathbf{W} \sim W_m(n, \mathbf{\Sigma}_w)$ (m-dimensional Wishart distribution with n degrees of freedom and covariance matrix $\mathbf{\Sigma}_w$), $\mathbf{z} \sim N_m(\boldsymbol{\mu}, \mathbf{\Sigma}_z)$, and they are independent of each other. Throughout the paper, $\mathbf{\Sigma}_w$ and $\mathbf{\Sigma}_z$ are assumed to be positive definite, while the condition $n > m-1$ is imposed to guarantee the non-singularity of $\mathbf{W}$. Following Muirhead (1982, p.87), the number of degrees of freedom n is allowed to be real.

In this section, we derive the density of $\mathbf{q} = \mathbf{Wz}$. This finding extends the results of Bodnar et al. (2013) and Yonenaga and Suzukawa (2023) who express the density of $\mathbf{L}'\mathbf{Wz}$ with $\mathbf{L}$ an $m \times p$ deterministic matrix as a $(p+2)$-dimensional integral. However, the matrix $\mathbf{L}$ must have rank $p < m$ in their setup and, as a result, these findings cannot be used to obtain the density of $\mathbf{q}$. In this section, we show that the density of $\mathbf{q}$ can be expressed as at most a double integral.

As an important special case, we consider $m = 1$, $n > 0$ and let $\mathbf{\Sigma}_w = \sigma_w^2$ and $\mathbf{\Sigma}_z = \sigma_z^2$. Then, $w \overset{d}{=} \sigma_w^2 v$ with $v \sim \chi_n^2$ and the density of $q = wz$ with $z \sim N(\mu, \sigma_z^2)$ is expressed as

$$f(q) = \mathbb{E}[f(q|v)] = \int_0^\infty \frac{1}{\sqrt{2\pi}\sigma_z\sigma_w^2 v} \exp\left(-\frac{(q-\sigma_w^2 v\mu)^2}{2\sigma_z^2\sigma_w^4 v^2}\right) f_n(v)\mathrm{d}v, \tag{1}$$

where $f_n(\cdot)$ stands for the density function of a χ_n^2 random variable. This shows that when $m = 1$, the density function of q can be obtained with a single integral.[1]

Next, we generalize (1) when $m \geq 2$ and $n > m-1$. Let $\tilde{\mathbf{W}} = \mathbf{\Sigma}_w^{-\frac{1}{2}}\mathbf{W}\mathbf{\Sigma}_w^{-\frac{1}{2}} \sim W_m(n, \mathbf{I}_m)$. Then, conditionally on $\tilde{\mathbf{W}}$, we have

$$\mathbf{q} = \mathbf{\Sigma}_w^{\frac{1}{2}}\tilde{\mathbf{W}}\mathbf{\Sigma}_w^{\frac{1}{2}}\mathbf{z}|\tilde{\mathbf{W}} \sim N(\mathbf{\Sigma}_w^{\frac{1}{2}}\tilde{\mathbf{W}}\mathbf{\Sigma}_w^{\frac{1}{2}}\boldsymbol{\mu}, \mathbf{\Sigma}_w^{\frac{1}{2}}\tilde{\mathbf{W}}\mathbf{\Sigma}_w^{\frac{1}{2}}\mathbf{\Sigma}_z\mathbf{\Sigma}_w^{\frac{1}{2}}\tilde{\mathbf{W}}\mathbf{\Sigma}_w^{\frac{1}{2}}). \tag{2}$$

For a positive integer k, let $\phi_k(\cdot; \boldsymbol{\eta}, \mathbf{\Omega})$ denote the density of $N_k(\boldsymbol{\eta}, \mathbf{\Omega})$, and define

$$\hat{\mathbf{q}} = \mathbf{\Sigma}_w^{-\frac{1}{2}}\mathbf{q}, \quad \boldsymbol{\eta} = \mathbf{\Sigma}_w^{\frac{1}{2}}\boldsymbol{\mu}, \quad \tilde{\mathbf{\Sigma}} = \mathbf{\Sigma}_w^{\frac{1}{2}}\mathbf{\Sigma}_z\mathbf{\Sigma}_w^{\frac{1}{2}}. \tag{3}$$

Since (2) states that $\mathbf{q} \overset{d}{=} \mathbf{\Sigma}_w^{\frac{1}{2}}\tilde{\mathbf{W}}\mathbf{x}$ with $\mathbf{x} \sim N_m(\boldsymbol{\eta}, \tilde{\mathbf{\Sigma}})$ independent of $\tilde{\mathbf{W}}$, the conditional density of $\mathbf{q}$ is given by

$$f(\mathbf{q}|\tilde{\mathbf{W}}) = \frac{\phi_m(\tilde{\mathbf{W}}^{-1}\hat{\mathbf{q}};\, \boldsymbol{\eta},\, \tilde{\mathbf{\Sigma}})}{|\mathbf{\Sigma}_w|^{\frac{1}{2}}|\tilde{\mathbf{W}}|}. \tag{4}$$

It depends on $\tilde{\mathbf{W}}$ only through $|\tilde{\mathbf{W}}|$ and the vector $\tilde{\mathbf{W}}^{-1}\hat{\mathbf{q}}$, whose joint distribution is given in the following lemma. Its proof is given in Appendix A.1.

Lemma 2.1. *Let $\tilde{\mathbf{W}} \sim W_m(n, \mathbf{I}_m)$, $m \geq 2$, $n > m-1$, and let $\mathbf{q}$ be an m-dimensional deterministic vector. Then, jointly,*

$$|\tilde{\mathbf{W}}| \overset{d}{=} \prod_{i=1}^m v_i, \qquad \tilde{\mathbf{W}}^{-1}\mathbf{q} \overset{d}{=} \frac{1}{v_1}\left(\mathbf{q} + \frac{\boldsymbol{\xi}}{\sqrt{v_2}}\right), \tag{5}$$

where $v_i \sim \chi_{n-m+i}^2$, $i = 1, \ldots, m$, $\boldsymbol{\xi} \sim N_m(\mathbf{0}_m, (\mathbf{q}'\mathbf{q})\mathbf{I}_m - \mathbf{q}\mathbf{q}')$, and they are mutually independent.

Theorem 2.1 follows from (4) and Lemma 2.1.

Theorem 2.1. *Let $\mathbf{W} \sim W_m(n, \mathbf{\Sigma}_w)$, $m \geq 2$, $n > m-1$, $\mathbf{z} \sim N_m(\boldsymbol{\mu}, \mathbf{\Sigma}_z)$, and $\mathbf{W}$ and $\mathbf{z}$ are independent. Then, the density of $\mathbf{q} = \mathbf{Wz}$ is given by*

$$f(\mathbf{q}) = \frac{\Gamma(n-m+1)}{\Gamma(n-1)|\mathbf{\Sigma}_w|^{\frac{1}{2}}} \int_0^\infty \int_0^\infty \frac{1}{v_1 v_2}\, \phi_m\left(\frac{\hat{\mathbf{q}}}{v_1};\, \boldsymbol{\eta},\, \tilde{\mathbf{\Sigma}} + \frac{(\hat{\mathbf{q}}'\hat{\mathbf{q}})\mathbf{I}_m - \hat{\mathbf{q}}\hat{\mathbf{q}}'}{v_1^2 v_2}\right) f_{n-m+1}(v_1) f_{n-m+2}(v_2)\mathrm{d}v_2\mathrm{d}v_1, \tag{6}$$

[1] Expression (1) also holds at $q = 0$, where it gives $f(0) = e^{-\mu^2/(2\sigma_z^2)}/[\sqrt{2\pi}\,\sigma_z\sigma_w^2(n-2)]$ when $n > 2$ and $f(0) = \infty$ when $n \leq 2$.

where $\Gamma(\cdot)$ *is the gamma function,* $f_d(\cdot)$ *denotes the density of the* χ^2*-distribution with* d *degrees of freedom, and* $\hat{\mathbf{q}}$*,* $\boldsymbol{\eta}$*, and* $\tilde{\boldsymbol{\Sigma}}$ *are defined in* (3)*. In particular, at the origin,*

$$f(\mathbf{0}_m) = \frac{e^{-\frac{\boldsymbol{\mu}'\boldsymbol{\Sigma}_z^{-1}\boldsymbol{\mu}}{2}}}{(2\pi)^{\frac{m}{2}}|\boldsymbol{\Sigma}_w|^{\frac{1}{2}}|\boldsymbol{\Sigma}_z|^{\frac{1}{2}}}\frac{\Gamma(n-m-1)}{\Gamma(n-1)} \tag{7}$$

when $n > m+1$*, and* $f(\mathbf{0}_m) = \infty$ *when* $m-1 < n \le m+1$*.*

The point $\mathbf{q}$ enters (6) only through $\hat{\mathbf{q}}/v_1$ and the matrix $(\hat{\mathbf{q}}'\hat{\mathbf{q}})\mathbf{I}_m - \hat{\mathbf{q}}\hat{\mathbf{q}}'$, so that (6) holds for every $\mathbf{q}$, including the origin, where the integrand reduces to $\phi_m(\mathbf{0}_m;\boldsymbol{\eta},\tilde{\boldsymbol{\Sigma}})/(v_1v_2)$ and (6) gives (7).

Proof of Theorem 2.1. By Lemma 2.1, applied with $\mathbf{q}$ replaced by $\hat{\mathbf{q}}$, the vector $\tilde{\mathbf{W}}^{-1}\hat{\mathbf{q}}$ is, conditionally on v_1 and v_2, normal with mean $\hat{\mathbf{q}}/v_1$ and covariance matrix $[(\hat{\mathbf{q}}'\hat{\mathbf{q}})\mathbf{I}_m - \hat{\mathbf{q}}\hat{\mathbf{q}}']/(v_1^2v_2)$, and it is independent of $v_3,\dots,v_m$. Let $\mathbf{x} \sim N_m(\mathbf{a},\mathbf{A})$, where $\mathbf{A}$ may be singular, and let $\boldsymbol{\varepsilon} \sim N_m(\mathbf{0}_m,\tilde{\boldsymbol{\Sigma}})$ be independent of $\mathbf{x}$. Since $\phi_m(\mathbf{x};\boldsymbol{\eta},\tilde{\boldsymbol{\Sigma}}) = \phi_m(\boldsymbol{\eta};\mathbf{x},\tilde{\boldsymbol{\Sigma}})$ is the density of $\mathbf{x}+\boldsymbol{\varepsilon}$ at $\boldsymbol{\eta}$ conditionally on $\mathbf{x}$, and $\mathbf{x}+\boldsymbol{\varepsilon} \sim N_m(\mathbf{a},\tilde{\boldsymbol{\Sigma}}+\mathbf{A})$ with $\tilde{\boldsymbol{\Sigma}}+\mathbf{A}$ positive definite, we have

$$\mathbb{E}\left[\phi_m(\mathbf{x};\boldsymbol{\eta},\tilde{\boldsymbol{\Sigma}})\right] = \int_{\mathbb{R}^m}\phi_m(\boldsymbol{\eta};\mathbf{y},\tilde{\boldsymbol{\Sigma}})\mathrm{d}F_{\mathbf{x}}(\mathbf{y}) = \phi_m(\boldsymbol{\eta};\mathbf{a},\tilde{\boldsymbol{\Sigma}}+\mathbf{A}) = \phi_m(\mathbf{a};\boldsymbol{\eta},\tilde{\boldsymbol{\Sigma}}+\mathbf{A}), \tag{8}$$

where $F_{\mathbf{x}}$ is the distribution function of $\mathbf{x}$. Taking the expectation of (4), first over $\boldsymbol{\xi}$ and then over $v_3,\dots,v_m$, and using $\mathbb{E}(v_i^{-1}) = 1/(n-m+i-2)$, so that

$$\prod_{i=3}^{m}\mathbb{E}\left(v_i^{-1}\right) = \frac{1}{(n-m+1)\cdots(n-2)} = \frac{\Gamma(n-m+1)}{\Gamma(n-1)}, \tag{9}$$

we obtain (6). At $\mathbf{q} = \mathbf{0}_m$, the integrand of (6) is $\phi_m(\mathbf{0}_m;\boldsymbol{\eta},\tilde{\boldsymbol{\Sigma}})\,(v_1v_2)^{-1}f_{n-m+1}(v_1)f_{n-m+2}(v_2)$. Since $\mathbb{E}(v_1^{-1}) = \infty$ for $n \le m+1$, and

$$\mathbb{E}\left(v_1^{-1}\right)\mathbb{E}\left(v_2^{-1}\right) = \frac{1}{(n-m-1)(n-m)} = \frac{\Gamma(n-m-1)}{\Gamma(n-m+1)} \tag{10}$$

for $n > m+1$, (7) follows from $|\tilde{\boldsymbol{\Sigma}}| = |\boldsymbol{\Sigma}_w||\boldsymbol{\Sigma}_z|$ and $\boldsymbol{\eta}'\tilde{\boldsymbol{\Sigma}}^{-1}\boldsymbol{\eta} = \boldsymbol{\mu}'\boldsymbol{\Sigma}_z^{-1}\boldsymbol{\mu}$. □

The results of Theorem 2.1 show that the density $f(\mathbf{q})$ can be obtained by using a double integral when $m \ge 2$. This finding is general and it is independent of the product dimension m.

Remark 2.2. Let $\mathbf{q} \ne \mathbf{0}_m$, $r = (\hat{\mathbf{q}}'\hat{\mathbf{q}})^{\frac{1}{2}}$, and let $\mathbf{Q} = [\tilde{\mathbf{q}},\ \mathbf{Q}_1]$ be an $m \times m$ orthonormal matrix whose first column is $\tilde{\mathbf{q}} = \hat{\mathbf{q}}/r$. For computation, it is convenient to evaluate the normal density in (6) in the coordinates defined by $\mathbf{Q}$. Write

$$\mathbf{Q}'\tilde{\boldsymbol{\Sigma}}\mathbf{Q} = \begin{bmatrix} \omega_{11} & \boldsymbol{\omega}_{21}' \\ \boldsymbol{\omega}_{21} & \boldsymbol{\Omega}_{22} \end{bmatrix}, \qquad \mathbf{Q}'\boldsymbol{\eta} = \begin{bmatrix} \eta_1 \\ \boldsymbol{\eta}_2 \end{bmatrix}, \tag{11}$$

and let $\mathbf{PDP}'$, $\mathbf{D} = \mathrm{Diag}(d_1,\dots,d_{m-1})$, be the spectral decomposition of $\boldsymbol{\Omega}_{22} - \boldsymbol{\omega}_{21}\boldsymbol{\omega}_{21}'/\omega_{11}$. With $x = r/v_1$ and $a = x^2/v_2$, we have

$$\frac{\mathbf{Q}'\hat{\mathbf{q}}}{v_1} = \begin{bmatrix} x \\ \mathbf{0}_{m-1} \end{bmatrix}, \qquad \frac{\mathbf{Q}'\left[(\hat{\mathbf{q}}'\hat{\mathbf{q}})\mathbf{I}_m - \hat{\mathbf{q}}\hat{\mathbf{q}}'\right]\mathbf{Q}}{v_1^2v_2} = \begin{bmatrix} 0 & \mathbf{0}_{m-1}' \\ \mathbf{0}_{m-1} & a\mathbf{I}_{m-1} \end{bmatrix}, \tag{12}$$

and, with $\mathbf{g} = \mathbf{P}'[\boldsymbol{\eta}_2 + \boldsymbol{\omega}_{21}(x-\eta_1)/\omega_{11}]$, the normal density in (6) equals

$$\frac{1}{(2\pi)^{\frac{m}{2}}\omega_{11}^{\frac{1}{2}}\prod_{i=1}^{m-1}(d_i+a)^{\frac{1}{2}}}\exp\left(-\frac{(x-\eta_1)^2}{2\omega_{11}} - \frac{1}{2}\sum_{i=1}^{m-1}\frac{g_i^2}{d_i+a}\right). \tag{13}$$

The exponent is a sum of nonpositive terms rather than a difference of quadratic forms of order $\|\boldsymbol{\mu}\|^2$, so its rounding error is not amplified when $\boldsymbol{\mu}$ is large, and the integrand costs $O(m)$ operations once $\mathbf{P}$ and $\mathbf{D}$ are computed.

3. Linear combinations

In this section, we derive the density function of p linear combinations of the product of a Wishart matrix and a normal vector expressed as $\mathbf{q} = \mathbf{L}'\mathbf{W}\mathbf{z}$, where $\mathbf{L}$ is a $m \times p$ matrix of rank p. If $p = m$, then the application of Theorem 2.1 yields the density of $\mathbf{q}$ by noting that

$$\mathbf{q} = \mathbf{L}'\mathbf{W}\mathbf{L}\mathbf{L}^{-1}\mathbf{z} = \mathbf{W}^*\mathbf{z}^*, \tag{14}$$

where $\mathbf{W}^* \sim W_m(n, \boldsymbol{\Sigma}_w^*)$ and $\mathbf{z}^* \sim N_m(\boldsymbol{\mu}^*, \boldsymbol{\Sigma}_z^*)$ with $\boldsymbol{\Sigma}_w^* = \mathbf{L}'\boldsymbol{\Sigma}_w\mathbf{L}$, $\boldsymbol{\mu}^* = \mathbf{L}^{-1}\boldsymbol{\mu}$, and $\boldsymbol{\Sigma}_z^* = \mathbf{L}^{-1}\boldsymbol{\Sigma}_z(\mathbf{L}')^{-1}$. Hence, the density function of $\mathbf{q}$ is given by Theorem 2.1 with $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}_z$, and $\boldsymbol{\Sigma}_w$ replaced by $\boldsymbol{\mu}^*$, $\boldsymbol{\Sigma}_z^*$, and $\boldsymbol{\Sigma}_w^*$, respectively.

If $p < m$, then the results of Theorem 2.1 cannot be applied directly. The derivation of the results is based on the following lemma.

Lemma 3.1. *Let $\tilde{\mathbf{W}} \sim W_m(n, \mathbf{I}_m)$, $m \geq 2$, $n > m-1$, and let $\tilde{\mathbf{z}}$ be an m-dimensional random vector independent of $\tilde{\mathbf{W}}$. Then, jointly with $\tilde{\mathbf{z}}$,*

$$\tilde{\mathbf{W}}\tilde{\mathbf{z}} \overset{d}{=} u_1\tilde{\mathbf{z}} + \sqrt{u_1}\,\boldsymbol{\xi}, \tag{15}$$

where $u_1 \sim \chi_n^2$ is independent of $\tilde{\mathbf{z}}$ and, conditionally on $\tilde{\mathbf{z}}$ and u_1, $\boldsymbol{\xi} \sim N_m(\mathbf{0}_m, (\tilde{\mathbf{z}}'\tilde{\mathbf{z}})\mathbf{I}_m - \tilde{\mathbf{z}}\tilde{\mathbf{z}}')$.

Proof of Lemma 3.1. For $\tilde{\mathbf{z}} = \mathbf{0}_m$, both sides of (15) vanish. For $\tilde{\mathbf{z}} \neq \mathbf{0}_m$, let $\check{\mathbf{z}} = \tilde{\mathbf{z}}/(\tilde{\mathbf{z}}'\tilde{\mathbf{z}})^{\frac{1}{2}}$ and let $\mathbf{Q}$ be an $m \times (m-1)$ orthonormal matrix with its columns orthogonal to $\tilde{\mathbf{z}}$. Since $[\check{\mathbf{z}},\ \mathbf{Q}]$ is an orthonormal matrix, Theorem 3.2.5 in Muirhead (1982) yields that, conditionally on $\tilde{\mathbf{z}}$,

$$[\check{\mathbf{z}},\ \mathbf{Q}]'\tilde{\mathbf{W}}[\check{\mathbf{z}},\ \mathbf{Q}] = \begin{bmatrix} \check{\mathbf{z}}'\tilde{\mathbf{W}}\check{\mathbf{z}} & \check{\mathbf{z}}'\tilde{\mathbf{W}}\mathbf{Q} \\ \mathbf{Q}'\tilde{\mathbf{W}}\check{\mathbf{z}} & \mathbf{Q}'\tilde{\mathbf{W}}\mathbf{Q} \end{bmatrix} \tag{16}$$

is $W_m(n, \mathbf{I}_m)$, which does not depend on $\tilde{\mathbf{z}}$, so that it is independent of $\tilde{\mathbf{z}}$. Furthermore, Theorem 3.2.10 in Muirhead (1982) gives $u_1 \equiv \check{\mathbf{z}}'\tilde{\mathbf{W}}\check{\mathbf{z}} \sim \chi_n^2$ and $\mathbf{Q}'\tilde{\mathbf{W}}\check{\mathbf{z}}|u_1 \sim N_{m-1}(\mathbf{0}_{m-1}, u_1\mathbf{I}_{m-1})$. Since $\check{\mathbf{z}}\check{\mathbf{z}}' + \mathbf{Q}\mathbf{Q}' = \mathbf{I}_m$,

$$\tilde{\mathbf{W}}\tilde{\mathbf{z}} = (\tilde{\mathbf{z}}'\tilde{\mathbf{z}})^{\frac{1}{2}}\left(\check{\mathbf{z}}\check{\mathbf{z}}' + \mathbf{Q}\mathbf{Q}'\right)\tilde{\mathbf{W}}\check{\mathbf{z}} = u_1\tilde{\mathbf{z}} + (\tilde{\mathbf{z}}'\tilde{\mathbf{z}})^{\frac{1}{2}}\mathbf{Q}\,\mathbf{Q}'\tilde{\mathbf{W}}\check{\mathbf{z}}, \tag{17}$$

and, conditionally on $\tilde{\mathbf{z}}$ and u_1, the last term is $N_m(\mathbf{0}_m, u_1(\tilde{\mathbf{z}}'\tilde{\mathbf{z}})\mathbf{Q}\mathbf{Q}')$, where $(\tilde{\mathbf{z}}'\tilde{\mathbf{z}})\mathbf{Q}\mathbf{Q}' = (\tilde{\mathbf{z}}'\tilde{\mathbf{z}})\mathbf{I}_m - \tilde{\mathbf{z}}\tilde{\mathbf{z}}'$. □

We now consider the general case of $\mathbf{q} = \mathbf{L}'\mathbf{W}\mathbf{z}$, where $\mathbf{L}$ is an $m \times p$ matrix, with rank $p < m$. Define $\tilde{\mathbf{L}} = \boldsymbol{\Sigma}_w^{\frac{1}{2}}\mathbf{L}$, $\tilde{\mathbf{W}} = \boldsymbol{\Sigma}_w^{-\frac{1}{2}}\mathbf{W}\boldsymbol{\Sigma}_w^{-\frac{1}{2}} \sim W_m(n, \mathbf{I}_m)$ and $\tilde{\mathbf{z}} = \boldsymbol{\Sigma}_w^{\frac{1}{2}}\mathbf{z} \sim N_m(\boldsymbol{\Sigma}_w^{\frac{1}{2}}\boldsymbol{\mu}, \boldsymbol{\Sigma}_w^{\frac{1}{2}}\boldsymbol{\Sigma}_z\boldsymbol{\Sigma}_w^{\frac{1}{2}})$. Since $\mathbf{q} = \mathbf{L}'\mathbf{W}\mathbf{z} = \tilde{\mathbf{L}}'\tilde{\mathbf{W}}\tilde{\mathbf{z}}$, Lemma 3.1 gives

$$\mathbf{q} \overset{d}{=} u_1\tilde{\mathbf{L}}'\tilde{\mathbf{z}} + \sqrt{u_1}\,\tilde{\mathbf{L}}'\boldsymbol{\xi}. \tag{18}$$

Conditionally on $\tilde{\mathbf{z}}$ and u_1, $\tilde{\mathbf{L}}'\boldsymbol{\xi} \sim N_p(\mathbf{0}_p, (\tilde{\mathbf{z}}'\tilde{\mathbf{z}})\mathbf{A})$, where

$$\mathbf{A} = \tilde{\mathbf{L}}'\left(\mathbf{I}_m - \frac{\tilde{\mathbf{z}}\tilde{\mathbf{z}}'}{\tilde{\mathbf{z}}'\tilde{\mathbf{z}}}\right)\tilde{\mathbf{L}} = (\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{\frac{1}{2}}\left(\mathbf{I}_p - \frac{\hat{\mathbf{z}}\hat{\mathbf{z}}'}{\hat{\mathbf{z}}'\hat{\mathbf{z}} + \hat{u}}\right)(\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{\frac{1}{2}} \tag{19}$$

is positive definite with probability one when $p < m$, and

$$\hat{\mathbf{z}} = (\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-\frac{1}{2}}\tilde{\mathbf{L}}'\tilde{\mathbf{z}}, \tag{20}$$

$$\hat{u} = \tilde{\mathbf{z}}'(\mathbf{I}_m - \tilde{\mathbf{L}}(\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-1}\tilde{\mathbf{L}}')\tilde{\mathbf{z}} = \tilde{\mathbf{z}}'\tilde{\mathbf{z}} - \hat{\mathbf{z}}'\hat{\mathbf{z}}. \tag{21}$$

The application of the stochastic representation (18) yields the density function of $\mathbf{q}$, conditional on $\hat{\mathbf{z}}$, $\hat{u}$ and u_1, expressed as

$$f(\mathbf{q}|\hat{\mathbf{z}}, \hat{u}, u_1) = \frac{1}{(2\pi)^{\frac{p}{2}}|\boldsymbol{\Omega}|^{\frac{1}{2}}}\exp\left(-\frac{(\mathbf{q}-\boldsymbol{\eta})'\boldsymbol{\Omega}^{-1}(\mathbf{q}-\boldsymbol{\eta})}{2}\right), \tag{22}$$

where $\boldsymbol{\eta} = u_1(\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{\frac{1}{2}}\hat{\mathbf{z}}$ and $\boldsymbol{\Omega} = u_1(\tilde{\mathbf{z}}'\tilde{\mathbf{z}})\mathbf{A}$, which depends on $\hat{\mathbf{z}}$, $\hat{u}$, and u_1 through only four variables: $\mathbf{q}'\boldsymbol{\Omega}^{-1}\mathbf{q}$, $\mathbf{q}'\boldsymbol{\Omega}^{-1}\boldsymbol{\eta}$, $\boldsymbol{\eta}'\boldsymbol{\Omega}^{-1}\boldsymbol{\eta}$, and $|\boldsymbol{\Omega}|$.

From the definition of $\boldsymbol{\Omega}$ and (21), we get

$$\boldsymbol{\Omega}^{-1} = \frac{1}{u_1(\tilde{\mathbf{z}}'\tilde{\mathbf{z}})}\mathbf{A}^{-1} = \frac{1}{u_1(\hat{\mathbf{z}}'\hat{\mathbf{z}}+\hat{u})}(\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-\frac{1}{2}}\left(\mathbf{I}_p + \frac{\hat{\mathbf{z}}\hat{\mathbf{z}}'}{\hat{u}}\right)(\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-\frac{1}{2}}, \tag{23}$$

$$|\boldsymbol{\Omega}| = u_1^p(\hat{\mathbf{z}}'\hat{\mathbf{z}}+\hat{u})^p\frac{|\tilde{\mathbf{L}}'\tilde{\mathbf{L}}|\hat{u}}{\hat{u}+\hat{\mathbf{z}}'\hat{\mathbf{z}}} = u_1^p(\hat{\mathbf{z}}'\hat{\mathbf{z}}+\hat{u})^{p-1}|\tilde{\mathbf{L}}'\tilde{\mathbf{L}}|\hat{u}. \tag{24}$$

Let

$$\mathbf{PDP}' = \boldsymbol{\Sigma}_z^{\frac{1}{2}}\boldsymbol{\Sigma}_w^{\frac{1}{2}}(\mathbf{I}_m - \tilde{\mathbf{L}}(\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-1}\tilde{\mathbf{L}}')\boldsymbol{\Sigma}_w^{\frac{1}{2}}\boldsymbol{\Sigma}_z^{\frac{1}{2}}, \tag{25}$$

where $\mathbf{D} = \mathrm{Diag}(d_1, \ldots, d_m)$ are the eigenvalues in descending order (with $d_{m-p+1} = \ldots = d_m = 0$) and the columns of $\mathbf{P}$ are the corresponding eigenvectors. Define $\hat{\mathbf{q}} = (\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-\frac{1}{2}}\mathbf{q}$, $\hat{\mathbf{L}} = \mathbf{P}'\boldsymbol{\Sigma}_z^{\frac{1}{2}}\boldsymbol{\Sigma}_w^{\frac{1}{2}}\tilde{\mathbf{L}}(\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-\frac{1}{2}}$, and

$$\mathbf{y} = \mathbf{P}'\boldsymbol{\Sigma}_z^{-\frac{1}{2}}\mathbf{z} \sim N(\boldsymbol{\mu}_y, \mathbf{I}_m), \quad \boldsymbol{\mu}_y = \mathbf{P}'\boldsymbol{\Sigma}_z^{-\frac{1}{2}}\boldsymbol{\mu}. \tag{26}$$

Then, it holds that

$$\begin{aligned}\hat{u} &= \tilde{\mathbf{z}}'(\mathbf{I}_m - \tilde{\mathbf{L}}(\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-1}\tilde{\mathbf{L}}')\tilde{\mathbf{z}} = \mathbf{z}'\boldsymbol{\Sigma}_z^{-\frac{1}{2}}\boldsymbol{\Sigma}_z^{\frac{1}{2}}\boldsymbol{\Sigma}_w^{\frac{1}{2}}(\mathbf{I}_m - \tilde{\mathbf{L}}(\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-1}\tilde{\mathbf{L}}')\boldsymbol{\Sigma}_w^{\frac{1}{2}}\boldsymbol{\Sigma}_z^{\frac{1}{2}}\boldsymbol{\Sigma}_z^{-\frac{1}{2}}\mathbf{z} \\ &= \mathbf{z}'\boldsymbol{\Sigma}_z^{-\frac{1}{2}}\mathbf{PDP}'\boldsymbol{\Sigma}_z^{-\frac{1}{2}}\mathbf{z} = \mathbf{y}'\mathbf{D}\mathbf{y} = \sum_{i=1}^{m-p} d_i y_i^2,\end{aligned} \tag{27}$$

$$\hat{\mathbf{z}} = (\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-\frac{1}{2}}\tilde{\mathbf{L}}'\tilde{\mathbf{z}} = (\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-\frac{1}{2}}\tilde{\mathbf{L}}'\boldsymbol{\Sigma}_w^{\frac{1}{2}}\boldsymbol{\Sigma}_z^{\frac{1}{2}}\mathbf{PP}'\boldsymbol{\Sigma}_z^{-\frac{1}{2}}\boldsymbol{\Sigma}_w^{-\frac{1}{2}}\tilde{\mathbf{z}} = (\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-\frac{1}{2}}\tilde{\mathbf{L}}'\boldsymbol{\Sigma}_w^{\frac{1}{2}}\boldsymbol{\Sigma}_z^{\frac{1}{2}}\mathbf{P}\mathbf{y} = \hat{\mathbf{L}}'\mathbf{y}. \tag{28}$$

Hence,

$$\theta_1 \equiv \mathbf{q}'\boldsymbol{\Omega}^{-1}\mathbf{q} = \frac{1}{u_1(\hat{\mathbf{z}}'\hat{\mathbf{z}}+\hat{u})}\left(\hat{\mathbf{q}}'\hat{\mathbf{q}} + \frac{(\hat{\mathbf{q}}'\hat{\mathbf{z}})^2}{\hat{u}}\right), \tag{29}$$

$$\theta_2 \equiv \mathbf{q}'\boldsymbol{\Omega}^{-1}\boldsymbol{\eta} = \frac{1}{u_1(\hat{\mathbf{z}}'\hat{\mathbf{z}}+\hat{u})}\left(u_1\hat{\mathbf{q}}'\hat{\mathbf{z}} + \frac{u_1\hat{\mathbf{q}}'\hat{\mathbf{z}}(\hat{\mathbf{z}}'\hat{\mathbf{z}})}{\hat{u}}\right) = \frac{\hat{\mathbf{q}}'\hat{\mathbf{z}}}{\hat{u}}, \tag{30}$$

$$\theta_3 \equiv \boldsymbol{\eta}'\boldsymbol{\Omega}^{-1}\boldsymbol{\eta} = \frac{1}{u_1(\hat{\mathbf{z}}'\hat{\mathbf{z}}+\hat{u})}u_1^2\hat{\mathbf{z}}'\left(\mathbf{I}_p + \frac{\hat{\mathbf{z}}\hat{\mathbf{z}}'}{\hat{u}}\right)\hat{\mathbf{z}} = \frac{u_1(\hat{\mathbf{z}}'\hat{\mathbf{z}})}{\hat{u}}. \tag{31}$$

Note that θ_1, θ_2, θ_3, and $|\boldsymbol{\Omega}|$ are only functions of $v_1 \equiv \hat{\mathbf{q}}'\hat{\mathbf{z}}$, $v_2 \equiv \hat{\mathbf{z}}'\hat{\mathbf{z}}$, $\hat{u}$, and u_1. Thus,

$$f(\mathbf{q}|\hat{z}, \hat{u}, u_1) = f(q|v_1, v_2, \hat{u}, u_1) = \frac{1}{(2\pi)^{\frac{p}{2}}|\boldsymbol{\Omega}|^{\frac{1}{2}}}\exp\left(-\frac{\theta_1 - 2\theta_2 + \theta_3}{2}\right), \tag{32}$$

and, consequently,

$$f_q(\mathbf{q}) = \int_0^\infty\int_0^\infty\int_0^\infty\int_{-\infty}^\infty f(\mathbf{q}|v_1, v_2, \hat{u}, u_1)f_{v_1,v_2,\hat{u}}(v_1, v_2, \hat{u})f_n(u_1)\mathrm{d}v_1\mathrm{d}v_2\mathrm{d}\hat{u}\mathrm{d}u_1. \tag{33}$$

Here $f_{v_1,v_2,\hat{u}}$ is the joint density of $(v_1, v_2, \hat{u})$, which exists for $p \geq 2$. For $p = 1$, $v_2 = v_1^2/(\hat{\mathbf{q}}'\hat{\mathbf{q}})$, and (33) and (37) below are to be read as expectations, $f_q(\mathbf{q}) = \mathbb{E}[f(\mathbf{q}|v_1, v_2, \hat{u}, u_1)]$.

To reduce the above expression to a triple integral, we make use of the following identity

$$\int_0^\infty u_1^{-\frac{p}{2}}\exp\left(-\frac{a}{2u_1} - \frac{bu_1}{2}\right)f_n(u_1)\mathrm{d}u_1 = \frac{1}{2^{\frac{n-2}{2}}\Gamma\left(\frac{n}{2}\right)}\left(\frac{a}{1+b}\right)^{\frac{n-p}{4}}K_{\frac{n-p}{2}}\left(\sqrt{a(1+b)}\right) \tag{34}$$

for $a > 0$ and $1 + b > 0$, where $K_\nu(x)$ is the modified Bessel function of the second kind. When $a = 0$ and $n > p$, we have

$$\int_0^\infty u_1^{-\frac{p}{2}} \exp\left(-\frac{bu_1}{2}\right) f_n(u_1)\mathrm{d}u_1 = \frac{\Gamma\left(\frac{n-p}{2}\right)}{2^{\frac{p}{2}}\Gamma\left(\frac{n}{2}\right)(1+b)^{\frac{n-p}{2}}}. \tag{35}$$

Since u_1 is independent of $\tilde{\mathbf{z}}$ and, consequently, of v_1, v_2, and $\hat{u}$, the identity (34) yields

$$\begin{aligned} f(\mathbf{q}|v_1, v_2, \hat{u}) &= \int_0^\infty f(\mathbf{q}|v_1, v_2, \hat{u}, u_1) f_n(u_1)\mathrm{d}u_1 \\ &= \frac{\exp\left(\frac{\hat{\mathbf{q}}'\hat{\mathbf{z}}}{\hat{u}}\right)\left[(\hat{\mathbf{q}}'\hat{\mathbf{q}})\hat{u} + (\hat{\mathbf{q}}'\hat{\mathbf{z}})^2\right]^{\frac{n-p}{4}}}{2^{\frac{n-2}{2}}\Gamma\left(\frac{n}{2}\right)(2\pi)^{\frac{p}{2}}|\tilde{\mathbf{L}}'\tilde{\mathbf{L}}|^{\frac{1}{2}}\hat{u}^{\frac{1}{2}}(\hat{\mathbf{z}}'\hat{\mathbf{z}} + \hat{u})^{\frac{n-1}{2}}} K_{\frac{n-p}{2}}\left(\frac{\left[(\hat{\mathbf{q}}'\hat{\mathbf{q}})\hat{u} + (\hat{\mathbf{q}}'\hat{\mathbf{z}})^2\right]^{\frac{1}{2}}}{\hat{u}}\right) \end{aligned} \tag{36}$$

when $\mathbf{q} \neq \mathbf{0}_p$. As a result, we have

$$f_q(\mathbf{q}) = \int_0^\infty \int_0^\infty \int_{-\infty}^\infty f(\mathbf{q}|v_1, v_2, \hat{u}) f_{v_1,v_2,\hat{u}}(v_1, v_2, \hat{u})\mathrm{d}v_1\mathrm{d}v_2\mathrm{d}\hat{u} \quad \text{for } \mathbf{q} \neq \mathbf{0}_p. \tag{37}$$

When $\mathbf{q} = \mathbf{0}_p$, we have $v_1 = 0$ and

$$f(\mathbf{0}_p|v_2, \hat{u}) = \int_0^\infty f(q|v_2, \hat{u}, u_1) f_n(u_1)\mathrm{d}u_1 = \frac{\Gamma\left(\frac{n-p}{2}\right)\hat{u}^{\frac{n-p-1}{2}}}{2^p\pi^{\frac{p}{2}}\Gamma\left(\frac{n}{2}\right)|\tilde{\mathbf{L}}'\tilde{\mathbf{L}}|^{\frac{1}{2}}(v_2 + \hat{u})^{\frac{n-1}{2}}}. \tag{38}$$

As a result, we have

$$f_q(\mathbf{0}_p) = \int_0^\infty \int_0^\infty f(\mathbf{0}_p|v_2, \hat{u}) f_{v_2,\hat{u}}(v_2, \hat{u})\mathrm{d}v_2\mathrm{d}\hat{u} = \frac{\Gamma\left(\frac{n-p}{2}\right)}{2^p\pi^{\frac{p}{2}}\Gamma\left(\frac{n}{2}\right)|\tilde{\mathbf{L}}'\tilde{\mathbf{L}}|^{\frac{1}{2}}} E\left[\frac{\hat{u}^{\frac{n-p-1}{2}}}{(v_2 + \hat{u})^{\frac{n-1}{2}}}\right]. \tag{39}$$

This is the expectation of a ratio of two quadratic forms in multivariate normal random variables, raised to different powers. Using the method in Bao and Kan (2013), this expectation can be computed using a single or a double integral, depending on whether $(n - p - 1)/2$ is an integer or not.

Finally, we note that the evaluation of the density $f_{v_1,v_2,\hat{u}}(v_1, v_2, \hat{u})$ is related to the derivation of joint density function of linear and quadratic forms in normally distributed random variables (see, e.g., Mathai and Provost, 1992; Schöne and Schmid, 2000; Bodnar et al., 2026). This joint density is not available in closed form, so that (37) is, in effect, an integral of dimension larger than three. In Section 4 we present an alternative representation based on the characteristic function of $\mathbf{q}$, in which the expectation over the normal vector is carried out analytically and the density is obtained as a $(p + 1)$-dimensional integral for every $p < m$. Next, we present the results in several important special cases: $p = 1$ in Section 3.1 and $p = 2$ in Section 3.2, in which the joint density $f_{v_1,v_2,\hat{u}}$ reduces to the conditional density of a single quadratic form given one or two linear forms.

3.1. Special case: One linear combination

When $p = 1$ and $q = \boldsymbol{\alpha}'\mathbf{W}\mathbf{z}$, we have $v_1 = \hat{q}\hat{z}$ and $v_2 = \hat{z}^2$. By writing $\mathbf{L} = \boldsymbol{\alpha}$, the triple integral (37) is reduced to the following double integral for $q \neq 0$

$$f_q(q) = \int_{-\infty}^\infty \int_0^\infty \frac{|\hat{q}|^{\frac{n-1}{2}}\exp\left(\frac{\hat{q}\hat{z}}{\hat{u}}\right)}{2^{\frac{n-1}{2}}\Gamma\left(\frac{n}{2}\right)\sqrt{\pi}(\alpha'\Sigma_w\alpha)^{\frac{1}{2}}\hat{u}^{\frac{1}{2}}(\hat{z}^2 + \hat{u})^{\frac{n-1}{4}}} K_{\frac{n-1}{2}}\left(\frac{|\hat{q}|(\hat{z}^2 + \hat{u})^{\frac{1}{2}}}{\hat{u}}\right) f_{\hat{u},\hat{z}}(\hat{u}, \hat{z})\mathrm{d}\hat{u}\mathrm{d}\hat{z}. \tag{40}$$

When $q = 0$, we can use (39) to obtain

$$f_q(0) = \frac{\Gamma\left(\frac{n-1}{2}\right)}{2\Gamma\left(\frac{n}{2}\right)\sqrt{\pi}(\alpha'\Sigma_w\alpha)^{\frac{1}{2}}} E\left[\frac{\hat{u}^{\frac{n-2}{2}}}{(\hat{z}^2+\hat{u})^{\frac{n-1}{2}}}\right]. \tag{41}$$

To evaluate the density of q using the above expression, we need the joint density of $(\hat{u}, \hat{z})$. Note that

$$\hat{z} \sim N\left(\frac{\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\mu}}{(\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\alpha})^{\frac{1}{2}}}, \frac{\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\Sigma}_z\boldsymbol{\Sigma}_w\boldsymbol{\alpha}}{\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\alpha}}\right). \tag{42}$$

Next, we show that conditional on $\hat{z}$, $\hat{u}$ is a quadratic form in multivariate normal random variables.

Let $\hat{\boldsymbol{\alpha}} = \mathbf{P}'\boldsymbol{\Sigma}_z^{\frac{1}{2}}\boldsymbol{\Sigma}_w\boldsymbol{\alpha}$ and let $\mathbf{Q} = [\hat{\boldsymbol{\alpha}}/(\hat{\boldsymbol{\alpha}}'\hat{\boldsymbol{\alpha}})^{\frac{1}{2}},\ \mathbf{Q}_1]$ be an $m\times m$ orthonormal matrix. Define $\mathbf{x} = \mathbf{Q}'\mathbf{y} \sim N_m(\mathbf{Q}'\boldsymbol{\mu}_y, \mathbf{I}_m)$. Then, it holds that

$$\hat{z} = \frac{\hat{\boldsymbol{\alpha}}'\mathbf{y}}{(\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\alpha})^{\frac{1}{2}}} = \left(\frac{\hat{\boldsymbol{\alpha}}'\hat{\boldsymbol{\alpha}}}{\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\alpha}}\right)^{\frac{1}{2}} x_1 = \left(\frac{\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\Sigma}_z\boldsymbol{\Sigma}_w\boldsymbol{\alpha}}{\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\alpha}}\right)^{\frac{1}{2}} x_1, \tag{43}$$

$$\hat{u} = \mathbf{y}'\mathbf{D}\mathbf{y} = \mathbf{y}'\mathbf{Q}\mathbf{Q}'\mathbf{D}\mathbf{Q}\mathbf{Q}'\mathbf{y} = \mathbf{x}'\mathbf{C}\mathbf{x}, \tag{44}$$

where

$$\mathbf{Q}'\mathbf{D}\mathbf{Q} = \mathbf{C} = \begin{bmatrix} C_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{21} & \mathbf{C}_{22} \end{bmatrix}, \tag{45}$$

with C_{11} the (1,1) element of $\mathbf{C}$. Denoting $\tilde{\mathbf{x}} = [x_2, \ldots, x_m]'$, we can then write

$$\hat{u} = C_{11}x_1^2 + 2x_1\mathbf{C}_{21}'\tilde{\mathbf{x}} + \tilde{\mathbf{x}}'\mathbf{C}_{22}\tilde{\mathbf{x}} = C_{11}\frac{\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\alpha}}{\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\Sigma}_z\boldsymbol{\Sigma}_w\boldsymbol{\alpha}}\hat{z}^2 + 2\left(\frac{\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\alpha}}{\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\Sigma}_z\boldsymbol{\Sigma}_w\boldsymbol{\alpha}}\right)^{\frac{1}{2}}\hat{z}\mathbf{C}_{21}'\tilde{\mathbf{x}} + \tilde{\mathbf{x}}'\mathbf{C}_{22}\tilde{\mathbf{x}}. \tag{46}$$

Hence, conditional on $\hat{z}$, $\hat{u}$ is a quadratic form in $\tilde{\mathbf{x}}$, and the evaluation of its density is discussed in Appendix A.2. Finally, the density of q is evaluated as

$$f_q(q) = \int_{-\infty}^{\infty}\int_{0}^{\infty} \frac{|\hat{q}|^{\frac{n-1}{2}}\exp\left(\frac{\hat{q}\hat{z}}{\hat{u}}\right)K_{\frac{n-1}{2}}\left(\frac{|\hat{q}|(\hat{z}^2+\hat{u})^{\frac{1}{2}}}{\hat{u}}\right)}{2^{\frac{n-1}{2}}\Gamma\left(\frac{n}{2}\right)\sqrt{\pi}(\alpha'\Sigma_w\alpha)^{\frac{1}{2}}\hat{u}^{\frac{1}{2}}(\hat{z}^2+\hat{u})^{\frac{n-1}{4}}} f_{\hat{u}|\hat{z}}(\hat{u})\mathrm{d}\hat{u} f_{\hat{z}}(\hat{z})\mathrm{d}\hat{z}, \quad q \neq 0. \tag{47}$$

Figure 1 plots the density of $q = \boldsymbol{\alpha}'\mathbf{W}\mathbf{z}$ for $m = 5$, $n \in \{9, 10\}$, and the parameters given in its caption. Both densities are unimodal and skewed to the right; when n is larger, the density shifts to the right and has a lower peak.

3.2. Special case: Two linear combinations

When $p = 2$, we have $v_1 = \hat{q}_1\hat{z}_1 + \hat{q}_2\hat{z}_2$ and $v_2 = \hat{z}_1^2 + \hat{z}_2^2$, where $\hat{\mathbf{z}} = \hat{\mathbf{L}}'\mathbf{y}$. Hence,

$$\begin{aligned} f_q(\mathbf{q}) &= \mathbb{E}[f_q(\mathbf{q}|v_1, v_2, \hat{u})] = \mathbb{E}[f_q(\mathbf{q}|\hat{z}_1, \hat{z}_2, \hat{u})] \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{0}^{\infty} \frac{\exp\left(\frac{v_1}{\hat{u}}\right)\left[(\hat{\mathbf{q}}'\hat{\mathbf{q}})\hat{u} + v_1^2\right]^{\frac{n-2}{4}}}{2^{\frac{n}{2}}\Gamma\left(\frac{n}{2}\right)\pi|\tilde{\mathbf{L}}'\tilde{\mathbf{L}}|^{\frac{1}{2}}\hat{u}^{\frac{1}{2}}(v_2+\hat{u})^{\frac{n-1}{2}}} K_{\frac{n-2}{2}}\left(\frac{\left[(\hat{\mathbf{q}}'\hat{\mathbf{q}})\hat{u} + v_1^2\right]^{\frac{1}{2}}}{\hat{u}}\right) f(\hat{u}, \hat{z}_1, \hat{z}_2)\mathrm{d}\hat{z}_1\mathrm{d}\hat{z}_2\mathrm{d}\hat{u}. \end{aligned} \tag{48}$$

Let $\mathbf{Q} = [\hat{\mathbf{L}}(\hat{\mathbf{L}}'\hat{\mathbf{L}})^{-\frac{1}{2}},\ \mathbf{Q}_1]$ be an $m \times m$ orthonormal matrix, where the columns of $\mathbf{Q}_1$ are orthogonal to $\hat{\mathbf{L}}$. Define $\mathbf{x} = \mathbf{Q}'\mathbf{y} \sim N_m(\mathbf{Q}'\boldsymbol{\mu}_y, \mathbf{I}_m)$. We can then write

$$\hat{\mathbf{z}} = \hat{\mathbf{L}}'\mathbf{y} = \hat{\mathbf{L}}'\mathbf{Q}\mathbf{Q}'\mathbf{y} = \hat{\mathbf{L}}'\mathbf{Q}\mathbf{x} = (\hat{\mathbf{L}}'\hat{\mathbf{L}})^{\frac{1}{2}}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \tag{49}$$

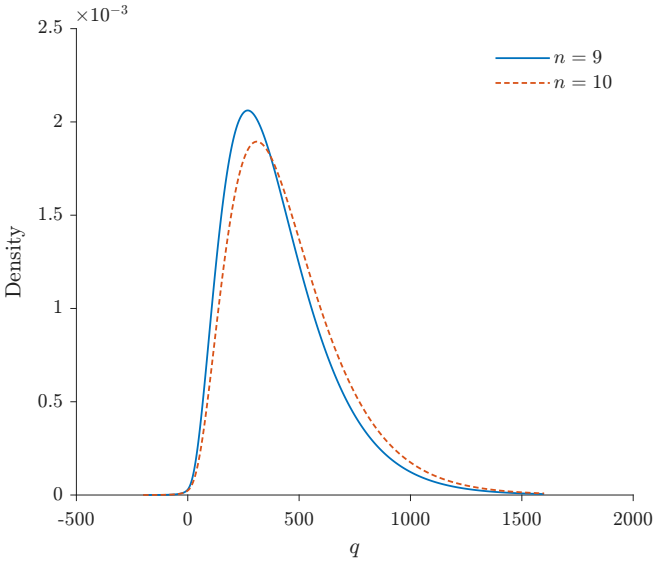


Figure 1: Density function of $q = \boldsymbol{\alpha}'\mathbf{W}\mathbf{z}$ for the cases of $n = 9$ and $n = 10$ when $\boldsymbol{\alpha} = \mathbf{1}_5$, $\boldsymbol{\mu} = [1, 2, 3, 4, 5]'$, $\boldsymbol{\Sigma}_w = 0.5\mathbf{I}_5 + 0.5\mathbf{1}_5\mathbf{1}_5'$, and $\boldsymbol{\Sigma}_z = 0.2\mathbf{I}_5 + 0.8\mathbf{1}_5\mathbf{1}_5'$.

$$\hat{u} = \mathbf{y}'\mathbf{D}\mathbf{y} = \mathbf{x}'\mathbf{Q}'\mathbf{D}\mathbf{Q}\mathbf{x} = \mathbf{x}'\mathbf{C}\mathbf{x}, \tag{50}$$

with

$$\mathbf{Q}'\mathbf{D}\mathbf{Q} \equiv \mathbf{C} = \begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{21} & \mathbf{C}_{22} \end{bmatrix}, \tag{51}$$

where $\mathbf{C}_{11}$ is 2×2. Conditional on $(\hat{z}_1, \hat{z}_2)$, or equivalently on (x_1, x_2), we can write $\hat{u}$ as a quadratic form in $\tilde{\mathbf{x}} = [x_3, \ldots, x_m]'$ as

$$\hat{u} = [x_1,\ x_2]\mathbf{C}_{11}[x_1,\ x_2]' + 2[x_1,\ x_2]\mathbf{C}_{21}'\tilde{\mathbf{x}} + \tilde{\mathbf{x}}'\mathbf{C}_{22}\tilde{\mathbf{x}}. \tag{52}$$

Since the relationship between $(\hat{z}_1, \hat{z}_2)$ and (x_1, x_2) is one-to-one, we get the density of $\mathbf{q}$ as in (48) conditionally on (x_1, x_2) where $f(\hat{u}, x_1, x_2) = f(\hat{u}|x_1, x_2)f(x_1)f(x_2)$, and the evaluation of $f(\hat{u}|x_1, x_2)$ is discussed in Appendix A.2.

3.3. Special case: $\boldsymbol{\Sigma}_z$ proportional to $\boldsymbol{\Sigma}_w^{-1}$

In this subsection, we assume that $\boldsymbol{\Sigma}_z = \sigma^2\boldsymbol{\Sigma}_w^{-1}$ for some $\sigma > 0$, which includes the case of identity covariance matrices, $\boldsymbol{\Sigma}_w = \boldsymbol{\Sigma}_z = \mathbf{I}_m$. Then $\tilde{\mathbf{z}} = \boldsymbol{\Sigma}_w^{\frac{1}{2}}\mathbf{z} \sim N_m(\tilde{\boldsymbol{\mu}}, \sigma^2\mathbf{I}_m)$ with $\tilde{\boldsymbol{\mu}} = \boldsymbol{\Sigma}_w^{\frac{1}{2}}\boldsymbol{\mu}$, and $\hat{\mathbf{z}}$ and $\hat{u}$ in (20) and (21) are independent with (see Theorem 5.1.3 and Theorem 5.5.1 in Mathai and Provost, 1992)

$$\hat{\mathbf{z}} \sim N_p(\hat{\boldsymbol{\mu}}, \sigma^2\mathbf{I}_p), \quad \hat{\boldsymbol{\mu}} = (\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-\frac{1}{2}}\tilde{\mathbf{L}}'\tilde{\boldsymbol{\mu}}, \tag{53}$$

$$\hat{u}/\sigma^2 \sim \chi^2_{m-p}(\hat{\delta}), \quad \hat{\delta} = (\tilde{\boldsymbol{\mu}}'\tilde{\boldsymbol{\mu}} - \hat{\boldsymbol{\mu}}'\hat{\boldsymbol{\mu}})/\sigma^2. \tag{54}$$

Let

$$w_1 = \frac{\hat{\mathbf{q}}'\hat{\mathbf{z}}}{\sqrt{\hat{\mathbf{q}}'\hat{\mathbf{q}}}} \quad \text{and} \quad \hat{u}_1 = \hat{\mathbf{z}}'\left(\mathbf{I}_p - \frac{\hat{\mathbf{q}}\hat{\mathbf{q}}'}{\hat{\mathbf{q}}'\hat{\mathbf{q}}}\right)\hat{\mathbf{z}}. \tag{55}$$

Then, w_1 and $\hat{u}_1$ are independent with $w_1 \sim N(\tilde{\mu}_1, \sigma^2)$, $\tilde{\mu}_1 = \hat{\mathbf{q}}'\hat{\boldsymbol{\mu}}/\sqrt{\hat{\mathbf{q}}'\hat{\mathbf{q}}}$, and $\hat{u}_1/\sigma^2 \sim \chi^2_{p-1}(\hat{\delta}_1)$, $\hat{\delta}_1 = (\hat{\boldsymbol{\mu}}'\hat{\boldsymbol{\mu}} - \tilde{\mu}_1^2)/\sigma^2$ (see Theorem 5.1.3 and Theorem 5.5.1 in Mathai and Provost, 1992). Moreover, $\hat{u}$, $\hat{u}_1$, and w_1 are mutually independent.

Since $v_1 = (\hat{\mathbf{q}}'\hat{\mathbf{q}})^{\frac{1}{2}} w_1$ and $v_2 = w_1^2 + \hat{u}_1$, the density (37) of $\mathbf{q} = \mathbf{L}'\mathbf{W}\mathbf{z}$ simplifies, for $p \geq 2$, to

$$f_q(\mathbf{q}) = \int_0^\infty \int_0^\infty \int_{-\infty}^\infty f(\mathbf{q}|w_1, \hat{u}_1, \hat{u})\, \phi_1(w_1; \tilde{\mu}_1, \sigma^2) f_{\hat{u}_1}(\hat{u}_1) f_{\hat{u}}(\hat{u}) \mathrm{d}w_1 \mathrm{d}\hat{u}_1 \mathrm{d}\hat{u} \quad \text{for } \mathbf{q} \neq \mathbf{0}_p, \tag{56}$$

where $f_{\hat{u}_1}$ and $f_{\hat{u}}$ are the density functions of $\hat{u}_1$ and $\hat{u}$, and

$$f(\mathbf{q}|w_1, \hat{u}_1, \hat{u}) = \frac{\exp\left(\frac{(\hat{\mathbf{q}}'\hat{\mathbf{q}})^{\frac{1}{2}} w_1}{\hat{u}}\right) \left[(\hat{\mathbf{q}}'\hat{\mathbf{q}})(\hat{u} + w_1^2)\right]^{\frac{n-p}{4}}}{2^{\frac{n-2}{2}} \Gamma\left(\frac{n}{2}\right) (2\pi)^{\frac{p}{2}} |\tilde{\mathbf{L}}'\tilde{\mathbf{L}}|^{\frac{1}{2}} \hat{u}^{\frac{1}{2}} (w_1^2 + \hat{u}_1 + \hat{u})^{\frac{n-1}{2}}} K_{\frac{n-p}{2}}\left(\frac{(\hat{\mathbf{q}}'\hat{\mathbf{q}})^{\frac{1}{2}} (\hat{u} + w_1^2)^{\frac{1}{2}}}{\hat{u}}\right). \tag{57}$$

When $\mathbf{q} = \mathbf{0}_p$, (39) applies, where now $v_2 = \hat{\mathbf{z}}'\hat{\mathbf{z}}$ and $\hat{u}$ are independent with $v_2/\sigma^2 \sim \chi^2_p(\hat{\boldsymbol{\mu}}'\hat{\boldsymbol{\mu}}/\sigma^2)$.

When $p = 1$ and $\mathbf{L} = \boldsymbol{\alpha}$, a simpler representation follows directly from (18). With $\tilde{\boldsymbol{\alpha}} = \boldsymbol{\Sigma}_w^{\frac{1}{2}} \boldsymbol{\alpha}$, we have $(\tilde{\mathbf{z}}'\tilde{\mathbf{z}})\mathbf{A} = (\tilde{\boldsymbol{\alpha}}'\tilde{\boldsymbol{\alpha}})\hat{u}$ by (19) and (21), so that

$$q \overset{d}{=} u_1 \tilde{\boldsymbol{\alpha}}'\tilde{\mathbf{z}} + \left[u_1 (\tilde{\boldsymbol{\alpha}}'\tilde{\boldsymbol{\alpha}})\hat{u}\right]^{\frac{1}{2}} x_0, \tag{58}$$

where $x_0 \sim N(0,1)$ is independent of u_1 and $\tilde{\mathbf{z}}$, and $\tilde{\boldsymbol{\alpha}}'\tilde{\mathbf{z}} \sim N(\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\mu}, \sigma^2\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\alpha})$ is independent of $\hat{u}$. Conditionally on u_1 and $\hat{u}$, q is therefore normally distributed with mean $u_1\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\mu}$ and variance $(\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\alpha})(\sigma^2 u_1^2 + u_1\hat{u})$, and

$$f_q(q) = \int_0^\infty \int_0^\infty \phi_1\left(q;\, u_1\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\mu},\, \sigma^2(\boldsymbol{\alpha}'\boldsymbol{\Sigma}_w\boldsymbol{\alpha})(u_1^2 + u_1 r)\right) f_{m-1}^{\hat{\delta}}(r) f_n(u_1) \mathrm{d}r \mathrm{d}u_1, \tag{59}$$

where $r = \hat{u}/\sigma^2$, $f_{m-1}^{\hat{\delta}}$ is the density function of $\chi^2_{m-1}(\hat{\delta})$, and f_n is the density function of χ^2_n. Unlike (56), (59) also holds at $q = 0$. In the setting of this section, the density can also be obtained from the characteristic function (Remark 4.5), as a double integral for $p = 1$ and a triple integral for $p \geq 2$, the same dimensions as (59) and (56). These integrals are nevertheless considerably faster to evaluate (Section 5.3), because the characteristic function is evaluated with fixed quadrature rules on a grid that is shared by all evaluation points, whereas (59) and (56) require nested adaptive quadrature at each point.

4. A characteristic function approach

The representations of Section 3 for $p < m$ integrate out the Wishart matrix analytically and leave an integral over the distribution of the normal vector, which enters through one linear form and two quadratic forms. In this section we reverse the order: we compute the characteristic function of $\mathbf{q} = \mathbf{L}'\mathbf{W}\mathbf{z}$, in which the expectation over the Wishart matrix is elementary and the expectation over the normal vector is a Gaussian integral, and then recover the density by Fourier inversion. The characteristic function turns out to be a one-dimensional integral for every $p \leq m$, and the density a $(p+1)$-dimensional integral, valid for arbitrary $\boldsymbol{\Sigma}_w$ and $\boldsymbol{\Sigma}_z$ and at every point of $\mathbb{R}^p$, including the origin, for every $p < m$, and for $p = m$ when $n > m + 1$. Throughout this section, $\tilde{\mathbf{L}} = \boldsymbol{\Sigma}_w^{\frac{1}{2}}\mathbf{L}$, $\tilde{\boldsymbol{\mu}} = \boldsymbol{\Sigma}_w^{\frac{1}{2}}\boldsymbol{\mu}$, and $\tilde{\boldsymbol{\Sigma}} = \boldsymbol{\Sigma}_w^{\frac{1}{2}}\boldsymbol{\Sigma}_z\boldsymbol{\Sigma}_w^{\frac{1}{2}}$ as before, so that $\tilde{\mathbf{z}} = \boldsymbol{\Sigma}_w^{\frac{1}{2}}\mathbf{z} \sim N_m(\tilde{\boldsymbol{\mu}}, \tilde{\boldsymbol{\Sigma}})$ and $\mathbf{q} = \tilde{\mathbf{L}}'\tilde{\mathbf{W}}\tilde{\mathbf{z}}$ with $\tilde{\mathbf{W}} \sim W_m(n, \mathbf{I}_m)$. For $\mathbf{t} \in \mathbb{R}^p$ we write

$$\mathbf{a} = \tilde{\mathbf{L}}\mathbf{t} \in \mathbb{R}^m, \qquad \mathbf{K}_\mathbf{a} = (\mathbf{a}'\mathbf{a})\mathbf{I}_m - \mathbf{a}\mathbf{a}', \tag{60}$$

and note that $\mathbf{K}_\mathbf{a}$ is positive semidefinite by the Cauchy–Schwarz inequality, with $\mathbf{K}_\mathbf{a}\mathbf{a} = \mathbf{0}_m$.

Theorem 4.1. *Let $\mathbf{W} \sim W_m(n, \boldsymbol{\Sigma}_w)$, $n > m - 1$, $\mathbf{z} \sim N_m(\boldsymbol{\mu}, \boldsymbol{\Sigma}_z)$, and $\mathbf{W}$ and $\mathbf{z}$ be independent, and let $\mathbf{L}$ be an $m \times p$ matrix of rank $p \leq m$. Then, the characteristic function of $\mathbf{q} = \mathbf{L}'\mathbf{W}\mathbf{z}$ is given by*

$$\psi(\mathbf{t}) = \mathbb{E}\left(e^{i\mathbf{t}'\mathbf{q}}\right) = e^{-\frac{\boldsymbol{\mu}'\boldsymbol{\Sigma}_z^{-1}\boldsymbol{\mu}}{2}} \int_0^\infty \left|\mathbf{I}_m + u\tilde{\boldsymbol{\Sigma}}\mathbf{K}_\mathbf{a}\right|^{-\frac{1}{2}} \exp\left(\frac{1}{2}\mathbf{r}_u'\left(\tilde{\boldsymbol{\Sigma}}^{-1} + u\mathbf{K}_\mathbf{a}\right)^{-1}\mathbf{r}_u\right) f_n(u)\mathrm{d}u \tag{61}$$

for $\mathbf{t} \in \mathbb{R}^p$*, where* $\mathbf{r}_u = \tilde{\boldsymbol{\Sigma}}^{-1}\tilde{\boldsymbol{\mu}} + iu\mathbf{a}$*,* $i = \sqrt{-1}$*, and* $f_n(\cdot)$ *is the density function of* χ^2_n*.*

Proof of Theorem 4.1. Let $\mathbf{a}_0 = \mathbf{L}\mathbf{t}$, so that $\mathbf{t}'\mathbf{q} = \mathbf{a}_0'\mathbf{W}\mathbf{z} = \operatorname{tr}(\mathbf{W}\mathbf{T})$ with $\mathbf{T} = \frac{1}{2}(\mathbf{z}\mathbf{a}_0' + \mathbf{a}_0\mathbf{z}')$ symmetric. Conditionally on $\mathbf{z}$, the characteristic function of the Wishart distribution (see, e.g., Muirhead (1982, Theorem 3.2.3)) yields

$$\mathbb{E}\left(e^{i\mathbf{t}'\mathbf{q}}\middle|\mathbf{z}\right) = \left|\mathbf{I}_m - 2i\boldsymbol{\Sigma}_w\mathbf{T}\right|^{-\frac{n}{2}} = \left|\mathbf{I}_m - i\boldsymbol{\Sigma}_w(\mathbf{z}\mathbf{a}_0' + \mathbf{a}_0\mathbf{z}')\right|^{-\frac{n}{2}}. \tag{62}$$

We can write $\mathbf{I}_m - i\boldsymbol{\Sigma}_w(\mathbf{z}\mathbf{a}_0' + \mathbf{a}_0\mathbf{z}') = \mathbf{I}_m + \mathbf{U}\mathbf{V}'$ with $\mathbf{U} = -i\boldsymbol{\Sigma}_w[\mathbf{z},\ \mathbf{a}_0]$ and $\mathbf{V} = [\mathbf{a}_0,\ \mathbf{z}]$, and the identity $|\mathbf{I}_m + \mathbf{U}\mathbf{V}'| = |\mathbf{I}_2 + \mathbf{V}'\mathbf{U}|$ gives

$$\mathbb{E}\left(e^{i\mathbf{t}'\mathbf{q}}\middle|\mathbf{z}\right) = \left[(1 - i\mathbf{a}_0'\boldsymbol{\Sigma}_w\mathbf{z})^2 + (\mathbf{a}_0'\boldsymbol{\Sigma}_w\mathbf{a}_0)(\mathbf{z}'\boldsymbol{\Sigma}_w\mathbf{z})\right]^{-\frac{n}{2}} = \left[1 - 2i\mathbf{a}'\tilde{\mathbf{z}} + \tilde{\mathbf{z}}'\mathbf{K}_\mathbf{a}\tilde{\mathbf{z}}\right]^{-\frac{n}{2}}, \tag{63}$$

where we used $\mathbf{a}_0'\boldsymbol{\Sigma}_w\mathbf{z} = \mathbf{a}'\tilde{\mathbf{z}}$, $\mathbf{a}_0'\boldsymbol{\Sigma}_w\mathbf{a}_0 = \mathbf{a}'\mathbf{a}$, $\mathbf{z}'\boldsymbol{\Sigma}_w\mathbf{z} = \tilde{\mathbf{z}}'\tilde{\mathbf{z}}$, and $(\mathbf{a}'\mathbf{a})(\tilde{\mathbf{z}}'\tilde{\mathbf{z}}) - (\mathbf{a}'\tilde{\mathbf{z}})^2 = \tilde{\mathbf{z}}'\mathbf{K}_\mathbf{a}\tilde{\mathbf{z}}$. The real part of the base in (63) equals $1 + \tilde{\mathbf{z}}'\mathbf{K}_\mathbf{a}\tilde{\mathbf{z}} \geq 1$, so the base lies in the open right half-plane and the power is taken with the principal branch. For any complex x with $\Re(x) > 0$, where $\Re(x)$ denotes the real part of x, we have the Gamma integral

$$x^{-\frac{n}{2}} = \frac{1}{\Gamma\left(\frac{n}{2}\right)}\int_0^\infty s^{\frac{n}{2}-1}e^{-sx}\mathrm{d}s = \int_0^\infty e^{-\frac{u(x-1)}{2}}f_n(u)\mathrm{d}u, \tag{64}$$

where the second equality follows from the substitution $s = u/2$. Applying this identity to (63) with $x - 1 = -2i\mathbf{a}'\tilde{\mathbf{z}} + \tilde{\mathbf{z}}'\mathbf{K}_\mathbf{a}\tilde{\mathbf{z}}$ yields

$$\mathbb{E}\left(e^{i\mathbf{t}'\mathbf{q}}\middle|\mathbf{z}\right) = \int_0^\infty \exp\left(iu\mathbf{a}'\tilde{\mathbf{z}} - \frac{u}{2}\tilde{\mathbf{z}}'\mathbf{K}_\mathbf{a}\tilde{\mathbf{z}}\right)f_n(u)\mathrm{d}u. \tag{65}$$

Taking the expectation over $\tilde{\mathbf{z}} \sim N_m(\tilde{\boldsymbol{\mu}}, \tilde{\boldsymbol{\Sigma}})$, interchanging the order of integration (the integrand in (65) is bounded by one in modulus), and completing the square in the Gaussian density (see, e.g., Mathai and Provost (1992, Chapter 3)), we obtain, for any symmetric positive semidefinite $\mathbf{M}$ and $\mathbf{c} \in \mathbb{R}^m$,

$$\mathbb{E}\left[\exp\left(-\frac{1}{2}\tilde{\mathbf{z}}'\mathbf{M}\tilde{\mathbf{z}} + i\mathbf{c}'\tilde{\mathbf{z}}\right)\right] = \left|\mathbf{I}_m + \tilde{\boldsymbol{\Sigma}}\mathbf{M}\right|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}\tilde{\boldsymbol{\mu}}'\tilde{\boldsymbol{\Sigma}}^{-1}\tilde{\boldsymbol{\mu}} + \frac{1}{2}\mathbf{r}'\left(\tilde{\boldsymbol{\Sigma}}^{-1} + \mathbf{M}\right)^{-1}\mathbf{r}\right), \tag{66}$$

with $\mathbf{r} = \tilde{\boldsymbol{\Sigma}}^{-1}\tilde{\boldsymbol{\mu}} + i\mathbf{c}$. Setting $\mathbf{M} = u\mathbf{K}_\mathbf{a}$ and $\mathbf{c} = u\mathbf{a}$ in (66), and noting that $\tilde{\boldsymbol{\mu}}'\tilde{\boldsymbol{\Sigma}}^{-1}\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu}'\boldsymbol{\Sigma}_z^{-1}\boldsymbol{\mu}$, completes the proof. □

Two features of Theorem 4.1 deserve comment. First, no restriction on p is needed: the result covers the full product ($p = m$, $\mathbf{L} = \mathbf{I}_m$), for which the characteristic function of $\mathbf{W}\mathbf{z}$ is a single integral for every m, as well as any number of linear combinations. Second, the proof does not rely on the stochastic representation of Lemma 3.1; it only uses the characteristic function of the Wishart distribution and a Gaussian integral. The χ^2_n variable u in (61) is nevertheless the same as u_1 in (18): conditionally on $(\tilde{\mathbf{z}}, u_1)$, $\mathbf{q}$ is normally distributed with mean $u_1\tilde{\mathbf{L}}'\tilde{\mathbf{z}}$ and covariance matrix $u_1(\tilde{\mathbf{z}}'\tilde{\mathbf{z}})\mathbf{A}$, and $\mathbf{t}'\mathbf{A}\mathbf{t} = \tilde{\mathbf{z}}'\mathbf{K}_\mathbf{a}\tilde{\mathbf{z}}/(\tilde{\mathbf{z}}'\tilde{\mathbf{z}})$, so that the integrand of (65) is precisely the conditional characteristic function of $\mathbf{q}$ given $(\tilde{\mathbf{z}}, u_1)$.

The next proposition bounds the rate of decay of ψ, which determines when the density can be recovered from (61) by Fourier inversion and also provides a truncation bound for its numerical evaluation. For a symmetric matrix $\mathbf{B}$, let $\lambda_{\min}(\mathbf{B})$ denote its smallest eigenvalue, and write $\lambda_{\min} = \lambda_{\min}(\tilde{\boldsymbol{\Sigma}})$.

Proposition 4.1. *Under the assumptions of Theorem 4.1, let* $c = \mathbf{a}'\mathbf{a} = \|\tilde{\mathbf{L}}\mathbf{t}\|^2$*. Then, for every* $\mathbf{t} \in \mathbb{R}^p$*,*

$$|\psi(\mathbf{t})| \leq \int_0^\infty (1 + uc\lambda_{\min})^{-\frac{m-1}{2}}e^{-\frac{\lambda_{\min}cu^2}{2}}f_n(u)\mathrm{d}u, \tag{67}$$

and, consequently, $|\psi(\mathbf{t})| = O\left(\|\tilde{\mathbf{L}}\mathbf{t}\|^{-\frac{n+m-1}{2}}\right)$ *as* $\|\mathbf{t}\| \to \infty$*. In particular,* ψ *is absolutely integrable on* $\mathbb{R}^p$ *whenever* $n > 2p - m + 1$*.*

Proof of Proposition 4.1. Fix $\mathbf{t}$ and $u > 0$, and write $\mathbf{M}_u = \tilde{\mathbf{\Sigma}}^{-1} + u\mathbf{K}_\mathbf{a}$ and $\mathbf{g} = \tilde{\mathbf{\Sigma}}^{-1}\tilde{\boldsymbol{\mu}}$, so that $\mathbf{r}_u = \mathbf{g} + iu\mathbf{a}$ and

$$\Re\left(\mathbf{r}_u'\mathbf{M}_u^{-1}\mathbf{r}_u\right) = \mathbf{g}'\mathbf{M}_u^{-1}\mathbf{g} - u^2\mathbf{a}'\mathbf{M}_u^{-1}\mathbf{a} \le \boldsymbol{\mu}'\mathbf{\Sigma}_z^{-1}\boldsymbol{\mu} - u^2\mathbf{a}'\mathbf{M}_u^{-1}\mathbf{a}, \tag{68}$$

where the inequality follows from $\mathbf{M}_u \succeq \tilde{\mathbf{\Sigma}}^{-1}$. Since $\tilde{\mathbf{\Sigma}}^{-1} \preceq \lambda_{\min}^{-1}\mathbf{I}_m$, we have $\mathbf{M}_u \preceq \lambda_{\min}^{-1}\mathbf{I}_m + u\mathbf{K}_\mathbf{a}$ and, because $\mathbf{K}_\mathbf{a}\mathbf{a} = \mathbf{0}_m$,

$$\mathbf{a}'\mathbf{M}_u^{-1}\mathbf{a} \ge \mathbf{a}'\left(\lambda_{\min}^{-1}\mathbf{I}_m + u\mathbf{K}_\mathbf{a}\right)^{-1}\mathbf{a} = \lambda_{\min}c. \tag{69}$$

Moreover, $\mathbf{K}_\mathbf{a} = c\mathbf{Q}_\mathbf{a}\mathbf{Q}_\mathbf{a}'$ for an $m \times (m-1)$ matrix $\mathbf{Q}_\mathbf{a}$ with orthonormal columns orthogonal to $\mathbf{a}$, so that

$$\left|\mathbf{I}_m + u\tilde{\mathbf{\Sigma}}\mathbf{K}_\mathbf{a}\right| = \left|\mathbf{I}_{m-1} + uc\mathbf{Q}_\mathbf{a}'\tilde{\mathbf{\Sigma}}\mathbf{Q}_\mathbf{a}\right| \ge (1 + uc\lambda_{\min})^{m-1}. \tag{70}$$

Using the two bounds in (61) proves (67). For the rate, use $f_n(u) \le u^{\frac{n}{2}-1}/[2^{\frac{n}{2}}\Gamma(\frac{n}{2})]$ and substitute $u = w/\sqrt{c}$ to obtain

$$|\psi(\mathbf{t})| \le \frac{c^{-\frac{n}{4}}}{2^{\frac{n}{2}}\Gamma\left(\frac{n}{2}\right)}\int_0^\infty \left(1 + w\sqrt{c}\lambda_{\min}\right)^{-\frac{m-1}{2}} e^{-\frac{\lambda_{\min}w^2}{2}} w^{\frac{n}{2}-1}\mathrm{d}w \le \frac{c^{-\frac{n+m-1}{4}}}{2^{\frac{n}{2}}\Gamma\left(\frac{n}{2}\right)\lambda_{\min}^{\frac{m-1}{2}}}\int_0^\infty w^{\frac{n-m+1}{2}-1}e^{-\frac{\lambda_{\min}w^2}{2}}\mathrm{d}w, \tag{71}$$

where the last integral is finite because $n > m-1$. Since $\|\tilde{\mathbf{L}}\mathbf{t}\|^2 \ge \lambda_{\min}(\tilde{\mathbf{L}}'\tilde{\mathbf{L}})\|\mathbf{t}\|^2$ with $\lambda_{\min}(\tilde{\mathbf{L}}'\tilde{\mathbf{L}}) > 0$, because $\tilde{\mathbf{L}}$ has rank p, and $|\psi(\mathbf{t})| \le 1$, it follows that ψ is integrable on $\mathbb{R}^p$ when $(n+m-1)/2 > p$. □

Corollary 4.1. *Under the assumptions of Theorem 4.1, if $n > 2p - m + 1$, then $\mathbf{q} = \mathbf{L}'\mathbf{W}\mathbf{z}$ has a bounded continuous density given, for every $\mathbf{q} \in \mathbb{R}^p$, by*

$$f_q(\mathbf{q}) = \frac{1}{(2\pi)^p}\int_{\mathbb{R}^p} \Re\left(e^{-i\mathbf{t}'\mathbf{q}}\psi(\mathbf{t})\right)\mathrm{d}\mathbf{t}, \tag{72}$$

with $\psi(\mathbf{t})$ given by (61). In particular, for $p = 1$ and $\mathbf{L} = \boldsymbol{\alpha}$,

$$f_q(q) = \frac{1}{\pi}\int_0^\infty \Re\left(e^{-itq}\psi(t)\right)\mathrm{d}t, \tag{73}$$

where, in (61), $\mathbf{a} = t\tilde{\boldsymbol{\alpha}}$ with $\tilde{\boldsymbol{\alpha}} = \mathbf{\Sigma}_w^{\frac{1}{2}}\boldsymbol{\alpha}$ and $\mathbf{K}_\mathbf{a} = t^2\left[(\tilde{\boldsymbol{\alpha}}'\tilde{\boldsymbol{\alpha}})\mathbf{I}_m - \tilde{\boldsymbol{\alpha}}\tilde{\boldsymbol{\alpha}}'\right]$.

Proof of Corollary 4.1. By Proposition 4.1, ψ is integrable, so the Fourier inversion theorem applies and (72) follows, the real part being justified by $\psi(-\mathbf{t}) = \overline{\psi(\mathbf{t})}$. Folding the integral in (72) onto $(0, \infty)$ by the same conjugate symmetry gives (73). □

Since $2p - m + 1 \le m - 1 < n$ when $p < m$, the condition $n > 2p - m + 1$ always holds for $p < m$; it restricts n only when $p = m$, where it reads $n > m + 1$, the condition under which the density of $\mathbf{W}\mathbf{z}$ is bounded (see (7)).

Representation (72) is a $(p+1)$-dimensional integral: a p-dimensional Fourier inversion with the one-dimensional integral (61) inside. In contrast to (37), it does not involve the joint density of the linear and quadratic forms v_1, v_2, and $\hat{u}$, which has been integrated out analytically in (66); nor does it require a separate treatment of $\mathbf{q} = \mathbf{0}_p$. For $p = 1$, (73) is a two-dimensional integral, as is the representation of Section 3.1, but the latter additionally requires the conditional density of $\hat{u}$ given $\hat{z}$, which is itself an infinite series or an inversion integral; for $p = 2$, (72) is a three-dimensional integral, while the representation of Section 3.2 is a three-dimensional integral with the conditional density of $\hat{u}$ given $(\hat{z}_1, \hat{z}_2)$ inside.

We conclude this section with four remarks: on the implementation of (61) and (72), which are used in the numerical results of Section 5, on the dimension of the alternative representations, and on a reduction of (72) when $\mathbf{\Sigma}_z$ is proportional to $\mathbf{\Sigma}_w^{-1}$.

Remark 4.2. The integrand of (61) can be evaluated with $O(m)$ operations per quadrature node. Let $\tilde{\mathbf{\Sigma}} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}'$ with $\mathbf{\Lambda} = \mathrm{Diag}(\lambda_1, \ldots, \lambda_m)$, $\mathbf{b} = \mathbf{V}'\mathbf{a}$, $c = \mathbf{a}'\mathbf{a} = \mathbf{b}'\mathbf{b} > 0$, $\mathbf{g} = \mathbf{\Lambda}^{-1}\mathbf{V}'\tilde{\boldsymbol{\mu}}$, $D_j = \lambda_j/(1 + uc\lambda_j)$, and $\delta_u = 1 - u\sum_{j=1}^m D_j b_j^2 = c^{-1}\sum_{j=1}^m b_j^2/(1 + uc\lambda_j) > 0$. Writing $\tilde{\mathbf{\Sigma}}^{-1} + u\mathbf{K}_\mathbf{a} = \mathbf{G}_u - u\mathbf{a}\mathbf{a}'$ with $\mathbf{G}_u = \tilde{\mathbf{\Sigma}}^{-1} + uc\mathbf{I}_m$, the matrix determinant lemma and the Sherman–Morrison formula give

$$\left|\mathbf{I}_m + u\tilde{\mathbf{\Sigma}}\mathbf{K}_\mathbf{a}\right| = \delta_u \prod_{j=1}^m (1 + uc\lambda_j), \tag{74}$$

$$\mathbf{r}_u' \left(\tilde{\mathbf{\Sigma}}^{-1} + u\mathbf{K}_\mathbf{a}\right)^{-1} \mathbf{r}_u = \sum_{j=1}^m D_j r_j^2 + \frac{u}{\delta_u}\left(\sum_{j=1}^m D_j r_j b_j\right)^2, \qquad r_j = g_j + iub_j. \tag{75}$$

The second expression for δ_u avoids the cancellation in $1 - u\sum_{j=1}^m D_j b_j^2$ when uc is large. The spectral decomposition of $\tilde{\mathbf{\Sigma}}$ is computed once; it does not depend on u, $\mathbf{t}$, or $\mathbf{q}$.

Remark 4.3. The characteristic function does not depend on $\mathbf{q}$. When the density is required at many points, for instance to plot it, (61) is evaluated once on a grid of frequencies and reused for every $\mathbf{q}$ in (72), so that a complete density curve costs little more than a single point. The envelope (67) provides an explicit bound for truncating the frequency domain.

Remark 4.4. When p is close to m, an alternative to (72) is to marginalize the density of Theorem 2.1. Let $\mathbf{L}_0$ be an $m \times (m - p)$ matrix with $\mathbf{L}_0'\mathbf{L} = \mathbf{0}_{(m-p)\times p}$ such that $\mathbf{H} = [\mathbf{L},\ \mathbf{L}_0]$ is nonsingular. Then $\mathbf{h} = \mathbf{H}'\mathbf{W}\mathbf{z} = \mathbf{W}^*\mathbf{z}^*$ with $\mathbf{W}^* \sim W_m(n, \mathbf{H}'\mathbf{\Sigma}_w\mathbf{H})$ and $\mathbf{z}^* \sim N_m(\mathbf{H}^{-1}\boldsymbol{\mu}, \mathbf{H}^{-1}\mathbf{\Sigma}_z(\mathbf{H}')^{-1})$ independent, its density $f_h(\cdot)$ is a double integral by Theorem 2.1, and

$$f_q(\mathbf{q}) = \int_{\mathbb{R}^{m-p}} f_h(\mathbf{q}, \mathbf{h}_2)\mathrm{d}\mathbf{h}_2, \tag{76}$$

an $(m - p + 2)$-dimensional integral. Combining (6), (72), and (76), the density of any p linear combinations of $\mathbf{W}\mathbf{z}$ can be computed from an integral of dimension at most $\min(p + 1,\ m - p + 2)$, which does not exceed four whenever $p \le 3$ or $p \ge m - 2$. Which route is faster in practice depends on p, m, and the covariance matrices; see Section 5.

Remark 4.5. Suppose that $\mathbf{\Sigma}_z = \sigma^2\mathbf{\Sigma}_w^{-1}$ for some $\sigma > 0$, as in Section 3.3. Then $\tilde{\mathbf{\Sigma}} = \sigma^2\mathbf{I}_m$, and $\mathbf{K}_\mathbf{a}$ has the eigenvalue $c = \mathbf{a}'\mathbf{a}$ with multiplicity $m - 1$ and the eigenvalue zero with eigenvector $\mathbf{a}$. Splitting $\mathbf{r}_u$ into its components along $\mathbf{a}$ and orthogonal to $\mathbf{a}$, (61) becomes $\psi(\mathbf{t}) = g(\mathbf{a}'\mathbf{a}, \mathbf{a}'\tilde{\boldsymbol{\mu}})$ with

$$g(C, B) = \int_0^\infty (1 + u\sigma^2 C)^{-\frac{m-1}{2}} \exp\left(\frac{u\left(B^2 - C\tilde{\boldsymbol{\mu}}'\tilde{\boldsymbol{\mu}}\right)}{2(1 + u\sigma^2 C)} + iuB - \frac{\sigma^2 u^2 C}{2}\right) f_n(u)\mathrm{d}u, \tag{77}$$

where the exponent has a nonpositive real part by the Cauchy–Schwarz inequality. Hence ψ depends on $\mathbf{t}$ only through $\|\tilde{\mathbf{L}}\mathbf{t}\|^2$ and $\mathbf{t}'\tilde{\mathbf{L}}'\tilde{\boldsymbol{\mu}}$. Let $\mathbf{s} = (\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{\frac{1}{2}}\mathbf{t}$, $\boldsymbol{\eta} = (\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-\frac{1}{2}}\tilde{\mathbf{L}}'\tilde{\boldsymbol{\mu}}$, $\mathbf{e} = \boldsymbol{\eta}/\|\boldsymbol{\eta}\|$ (any unit vector if $\boldsymbol{\eta} = \mathbf{0}_p$), $\hat{\mathbf{q}} = (\tilde{\mathbf{L}}'\tilde{\mathbf{L}})^{-\frac{1}{2}}\mathbf{q}$, $\hat{q}_1 = \mathbf{e}'\hat{\mathbf{q}}$, and $\kappa = \|\hat{\mathbf{q}} - \hat{q}_1\mathbf{e}\|$. Writing $\mathbf{s} = s_1\mathbf{e} + \mathbf{s}_\perp$ in (72), the integral over $\mathbf{s}_\perp \in \mathbb{R}^{p-1}$ is a Hankel transform of a radial function, and for $p \ge 2$

$$f_q(\mathbf{q}) = (2\pi)^{-\frac{p+1}{2}} |\tilde{\mathbf{L}}'\tilde{\mathbf{L}}|^{-\frac{1}{2}} \int_{-\infty}^\infty \int_0^\infty \Re\left(e^{-is_1\hat{q}_1} g\left(s_1^2 + \rho^2, \|\boldsymbol{\eta}\|s_1\right)\right) \rho^{p-2} \Lambda_{\frac{p-3}{2}}(\kappa\rho)\mathrm{d}\rho\mathrm{d}s_1, \tag{78}$$

where $\Lambda_\nu(x) = J_\nu(x)/x^\nu$, with $\Lambda_\nu(0) = 1/[2^\nu\Gamma(\nu + 1)]$, and $J_\nu(\cdot)$ is the Bessel function of the first kind of order ν; for $p = 1$, (73) applies with $\psi(t) = g(t^2\tilde{\boldsymbol{\alpha}}'\tilde{\boldsymbol{\alpha}}, t\tilde{\boldsymbol{\alpha}}'\tilde{\boldsymbol{\mu}})$. Together with (77), (78) expresses the density as a three-dimensional integral for every $2 \le p < m$, and (73) as a two-dimensional integral for $p = 1$. These are the same dimensions as the representations (56) and (59) of Section 3.3, but, as Section 5.3 shows, the characteristic function representation is considerably faster to evaluate.

5. Numerical evaluation and validation

In this section we validate the representations of Sections 2–4 against simulation and compare their computing times. All integrals are evaluated with adaptive quadrature (`integral`, `integral2`, and `integral3` in MATLAB), except for the characteristic function (61), whose integrand is evaluated on Gauss–Legendre nodes in $\log u$ over the effective support of f_n, their number being doubled from 128 until ψ changes by less than 10^{-12}, and for the Fourier inversion (72), which is computed by a Filon rule on a tensor-product grid. Along each coordinate t_j the grid consists of N_s nodes $t_{j,k} = T_j \sinh(\beta_j x_k)/\sinh(\beta_j)$, where $x_1, \dots, x_{N_s}$ are equally spaced on $[-1, 1]$, T_j is the frequency beyond which $|\psi| < 10^{-10}$ along the jth axis (the box is enlarged until $|\psi| \le 10^{-9}$ on its whole boundary), and β_j is set from the width of the main lobe of $|\psi|$, so that the nodes are concentrated where ψ varies. Writing $e^{-i\mathbf{t}'\mathbf{q}} = \prod_{j=1}^{p} e^{-it_j q_j}$, each factor is integrated exactly against the piecewise cubic interpolant of ψ in t_j, which gives weights $w_{j,k}(q_j)$ in closed form and

$$f_q(\mathbf{q}) \approx \frac{1}{(2\pi)^p} \Re\left(\sum_{k_1=1}^{N_s} \cdots \sum_{k_p=1}^{N_s} \left[\prod_{j=1}^{p} w_{j,k_j}(q_j) \right] \psi(t_{1,k_1}, \dots, t_{p,k_p}) \right), \tag{79}$$

a sum that is evaluated as a sequence of matrix products. The oscillation of $e^{-i\mathbf{t}'\mathbf{q}}$, which is rapid when $\|\mathbf{q}\|$ is large, is thus handled analytically and the grid only has to resolve ψ; the error of (79) decreases like N_s^{-4}, and we use $N_s = 3001$, 601, and 181 for $p = 1$, 2, and 3. The discretization error is monitored by repeating (79) on every other node, and the truncation error is bounded by integrating the envelope of Proposition 4.1 outside the box. By Proposition 4.1, $|\psi(\mathbf{t})|$ decays like $\|\mathbf{t}\|^{-\alpha}$ with $\alpha = (n + m - 1)/2$, so that the truncation error behaves like $T^{p-\alpha}$; when $\alpha - p < 2$, the inversion is repeated on a box four times larger and extrapolated in T. When $\mathbf{\Sigma}_z$ is proportional to $\mathbf{\Sigma}_w^{-1}$, we use (78) instead, with the same Filon rule in s_1 on 700 nodes and composite Gauss–Legendre rules in ρ on panels shorter than half a period of $\Lambda_{\frac{p-3}{2}}(\kappa\rho)$, for every p. Since ψ does not depend on $\mathbf{q}$, it is evaluated once on the grid and reused for every evaluation point (Remark 4.3). As the benchmark we use the unbiased conditional Monte Carlo estimator

$$\hat{f}_q(\mathbf{q}) = \frac{1}{N} \sum_{r=1}^{N} \phi_p\left(\mathbf{q};\, \mathbf{L}'\mathbf{W}_r\boldsymbol{\mu},\, \mathbf{L}'\mathbf{W}_r\mathbf{\Sigma}_z\mathbf{W}_r\mathbf{L}\right), \tag{80}$$

where ϕ_p is defined before (3), and $\mathbf{W}_1, \dots, \mathbf{W}_N$ are independent draws from $W_m(n, \mathbf{\Sigma}_w)$, generated by the Bartlett decomposition so that real n is allowed. Unlike a kernel density estimator, (80) is unbiased, and its standard error is available from the same draws; the standard error requires the summand of (80) to have a finite second moment, which at $\mathbf{q} = \mathbf{0}_m$ with $p = m$ holds only for $n > m + 3$.

5.1. The density of the product: a bivariate illustration

We consider $m = 2$, $n = 6$, $\boldsymbol{\mu} = [0.7,\ -0.4]'$, and the non-proportional covariance matrices

$$\mathbf{\Sigma}_w = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1.3 \end{bmatrix}, \qquad \mathbf{\Sigma}_z = \begin{bmatrix} 1.4 & -0.3 \\ -0.3 & 0.9 \end{bmatrix}. \tag{81}$$

Figure 2 overlays the contours of the density of Theorem 2.1, evaluated by numerical integration on a 61×61 grid that is refined to a spacing of 0.1 near the peak, with the contours of a bivariate Gaussian kernel density estimate computed from 10^6 simulated draws of $\mathbf{q} = \mathbf{W}\mathbf{z}$ with Silverman's bandwidth. The two sets of contours are visually indistinguishable, apart from the innermost contour, where the kernel estimate is flattened by the smoothing.

Since a kernel estimate is biased, Table 1 reports a pointwise comparison at six points of three independent computations of the same density: the double integral of Theorem 2.1, the three-dimensional characteristic function inversion of Corollary 4.1 (here $p = m = 2$, and $n = 6 > m + 1$ as required), and the conditional Monte Carlo estimator (80) with $N = 10^6$. The two analytic representations agree to at least five significant digits (the largest relative difference is 4×10^{-6}), and both agree with the simulation to within Monte Carlo error at every point, including the origin, where Theorem 2.1 gives the closed-form value (7). The kernel estimate is accurate to within 3% away from the origin, but it is 25% too low at the origin: the density has a sharp peak there (its value at the origin is twice its value at $E[\mathbf{q}]$, and it decreases by a quarter within a distance of four from the origin, which is less than a quarter of a standard deviation of either coordinate), and the peak is narrower than the bandwidth of the kernel. This is a feature of

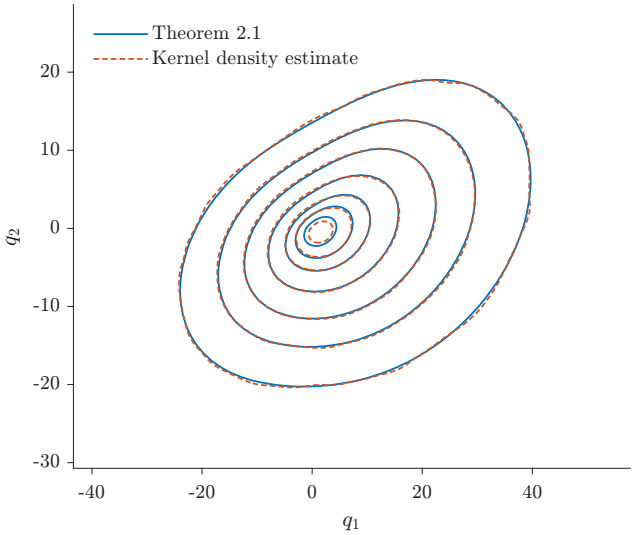


Figure 2: Contours of the density of $\mathbf{q} = \mathbf{W}\mathbf{z}$ for $m = 2$, $n = 6$, $\boldsymbol{\mu} = [0.7, -0.4]'$, and the covariance matrices in (81). Solid blue lines: Theorem 2.1, evaluated by numerical integration; dashed orange lines: bivariate kernel density estimate based on 10^6 simulated draws.

the family: the density of $\mathbf{q}$ is the mixture of the $N_m(\mathbf{W}\boldsymbol{\mu}, \mathbf{W}\boldsymbol{\Sigma}_z\mathbf{W})$ densities over the Wishart distribution of $\mathbf{W}$, and the components with $\mathbf{W}$ close to the zero matrix are concentrated near the origin; for $n \leq m + 1$ they make the peak unbounded. Kernel smoothing therefore cannot replace the exact density where it varies rapidly.

Table 1: Density of $\mathbf{q} = \mathbf{W}\mathbf{z}$ at selected points for the bivariate example of Figure 2: Theorem 2.1, characteristic function inversion (Corollary 4.1), kernel density estimate, and the conditional Monte Carlo estimator (80) with $N = 10^6$ (standard error in parentheses).

q_1	q_2	Theorem 2.1	Corollary 4.1	Kernel estimate	Monte Carlo
7.203	-1.040	0.00209913	0.00209913	0.00208970	0.00209808 (2.1147×10^{-6})
-9.656	8.854	9.6366×10^{-5}	9.6366×10^{-5}	9.6956×10^{-5}	9.6419×10^{-5} (1.8686×10^{-7})
24.061	-10.934	1.3222×10^{-4}	1.3222×10^{-4}	1.3608×10^{-4}	1.3193×10^{-4} (3.1944×10^{-7})
40.920	18.748	4.8098×10^{-5}	4.8098×10^{-5}	4.9268×10^{-5}	4.7863×10^{-5} (1.5861×10^{-7})
-18.085	-15.880	1.0118×10^{-4}	1.0118×10^{-4}	1.0313×10^{-4}	1.0112×10^{-4} (2.1001×10^{-7})
0	0	0.00421927	0.00421927	0.00317407	0.00420849 (8.4485×10^{-6})

5.2. Linear combinations: validation

Table 2: Analytic density of $\mathbf{q} = \mathbf{L}'\mathbf{W}\mathbf{z}$ versus the conditional Monte Carlo estimator (80) with $N = 10^6$, for $m = 5$, $\boldsymbol{\mu} = [1,\ 2,\ 3,\ 4,\ 5]'$, and $n = 10$ unless stated otherwise. General setting: $\boldsymbol{\Sigma}_w = 0.5\mathbf{I}_5 + 0.5\mathbf{1}_5\mathbf{1}_5'$, $\boldsymbol{\Sigma}_z = 0.2\mathbf{I}_5 + 0.8\mathbf{1}_5\mathbf{1}_5'$; identity setting: $\boldsymbol{\Sigma}_w = \boldsymbol{\Sigma}_z = \mathbf{I}_5$. The last column reports the absolute discrepancy in Monte Carlo standard errors.

Setting	Method	Analytic	Monte Carlo	|diff|/s.e.
General, $p = 1$	CF	0.00154694	$0.00154953\ (1.0619 \times 10^{-6})$	2.44
General, $p = 1$	Bessel	0.00154694	$0.00154953\ (1.0619 \times 10^{-6})$	2.44
General, $p = 2$	CF	1.9451×10^{-5}	$1.9537 \times 10^{-5}\ (7.1986 \times 10^{-8})$	1.20
General, $p = 3$	CF	2.8710×10^{-7}	$2.8943 \times 10^{-7}\ (3.8331 \times 10^{-9})$	0.61
General, $p = m - 1 = 4$	Marg.	6.5626×10^{-9}	$6.3308 \times 10^{-9}\ (3.1679 \times 10^{-10})$	0.73
General, $p = m = 5$	Thm 2.1	1.8893×10^{-10}	$1.5087 \times 10^{-10}\ (2.0361 \times 10^{-11})$	1.87
General, $p = 1$, $n = m - 0.5$	CF	0.00248545	$0.00249038\ (2.4319 \times 10^{-6})$	2.03
General, $p = 1$, $\mu = 0$	CF	0.00360820	$0.00360262\ (2.0322 \times 10^{-6})$	2.75
Identity, $p = 1$	Prop.	0.00512567	$0.00512119\ (5.6168 \times 10^{-6})$	0.80
Identity, $p = 2$	Prop.	6.6313×10^{-5}	$6.6335 \times 10^{-5}\ (1.3461 \times 10^{-7})$	0.17
Identity, $p = 2$	CF	6.6313×10^{-5}	$6.6335 \times 10^{-5}\ (1.3461 \times 10^{-7})$	0.17
General, $p = 1$, $q = 0$	CF	2.3920×10^{-5}	$2.3831 \times 10^{-5}\ (6.9960 \times 10^{-8})$	1.27

Table 2 compares the analytic density of $\mathbf{q} = \mathbf{L}'\mathbf{W}\mathbf{z}$ with the conditional Monte Carlo estimator (80) ($N = 10^6$) for $m = 5$, $\boldsymbol{\mu} = [1,\ 2,\ 3,\ 4,\ 5]'$, $n = 10$, and the general covariance matrices $\boldsymbol{\Sigma}_w = 0.5\mathbf{I}_5 + 0.5\mathbf{1}_5\mathbf{1}_5'$ and $\boldsymbol{\Sigma}_z = 0.2\mathbf{I}_5 + 0.8\mathbf{1}_5\mathbf{1}_5'$ of Figure 1, for p ranging from one to m, as well as for a value of n close to the boundary $n = m - 1$, a centered mean vector, identity covariance matrices (a special case of Section 3.3), and the origin. With $\mathbf{e}_j$ the jth column of $\mathbf{I}_5$, the matrices $\mathbf{L}$ for $p = 1, \dots, 5$ are

$$\mathbf{1}_5, \quad [\mathbf{e}_1 + \mathbf{e}_3 + \mathbf{e}_5,\ \mathbf{e}_2 + \mathbf{e}_3 + \mathbf{e}_4], \quad [\mathbf{e}_1 + \mathbf{e}_4,\ \mathbf{e}_2 + \mathbf{e}_4 + \mathbf{e}_5,\ \mathbf{e}_3 + \mathbf{e}_5], \quad [\mathbf{e}_1 + \mathbf{e}_5, \dots, \mathbf{e}_4 + \mathbf{e}_5], \quad \mathbf{I}_5, \tag{82}$$

and the evaluation point is $\mathbf{q} = E[\mathbf{q}] + \boldsymbol{\delta}$, with $E[\mathbf{q}] = n\mathbf{L}'\boldsymbol{\Sigma}_w\boldsymbol{\mu}$, $\boldsymbol{\delta} = 5$, $[3,\ -4]'$, $[3,\ -4,\ 2]'$, $[3,\ -4,\ 2,\ 1]'$, and $[3,\ -4,\ 2,\ 1,\ -2]'$ for $p = 1, \dots, 5$ in the general setting, $\boldsymbol{\delta} = 2$ for $n = m - 0.5$, $\boldsymbol{\delta} = 5$ for $\boldsymbol{\mu} = \mathbf{0}_5$, $\boldsymbol{\delta} = 2$ and $[1,\ -1]'$ in the identity setting, and $\mathbf{q} = 0$ in the last row. The column "Method" indicates the representation used: Theorem 2.1 for $p = m$ (Thm 2.1), Corollary 4.1 (CF), the Bessel-function representations of Sections 3.1–3.2 (Bessel), the representations of Section 3.3 for $\boldsymbol{\Sigma}_z \propto \boldsymbol{\Sigma}_w^{-1}$ (Prop.), and the marginalization (76) for $p = m - 1$ (Marg.). In every case the analytic value agrees with the simulation to within Monte Carlo error (the largest discrepancy is 2.75 standard errors, and the discrepancies are of both signs); the two analytic representations available for $p = 1$ in the general setting, and for $p = 2$ in the identity setting, agree with each other to all reported digits.

5.3. Computing time

Table 3 reports the time needed to evaluate the density at a single point and, for the characteristic function representation and Theorem 2.1, on a grid of 101 points, for several values of m and p, with $n = m + 5$, $\boldsymbol{\mu} = [1,\ 2, \dots, m]'/m$, $\mathbf{L} = [\mathbf{I}_p,\ \mathbf{1}_p\mathbf{1}_{m-p}'/p]'$ ($\mathbf{L} = \mathbf{I}_m$ for $p = m$), and the $m \times m$ covariance matrices of the general and identity settings of Table 2. The evaluation point is $E[\mathbf{q}] + \mathbf{s}/2$, where $\mathbf{s}$ is the vector of standard deviations of $\mathbf{q}$, and the grid is $E[\mathbf{q}] + t\mathbf{s}$ for $p = 1$ and $E[\mathbf{q}] + t\mathbf{s}/2$ for $p \geq 2$, with 101 equally spaced values of $t \in [-3, 3]$. Each time is the median of three runs (one run when it exceeds a minute), after an untimed warm-up of each representation, using MATLAB R2025b on a laptop computer with an Intel Core Ultra 9 185H processor. With the general covariance matrices, the characteristic function representation costs about 0.1 s per point for $p = 1$, about 5 s for $p = 2$, and 90–110 s for $p = 3$; in the identity setting, where the reduced representation of Remark 4.5 applies, it costs at most 2 s per point for every $p < m$ considered, up to $p = 9$ with $m = 10$. In both cases, a grid of 101 points costs at most a few times as much as a single point, because ψ is evaluated once (Remark 4.3). The Bessel-function representations of Sections 3.1–3.2 cost 0.05–0.09 s per point for $p = 1$ and 11–16 s for $p = 2$; at a single point they are competitive, but the series of Ruben (1962) has to be evaluated at every node of a nested adaptive quadrature, and nothing is shared across evaluation points, so that a grid of 101 points costs about 100 times as much. The representations of Section 3.3 cost between 0.2 and 17 s per point, and the marginalization (76) between 10 and 16 s per point for $p = m - 1$, which is much less than the characteristic function representation for $p = 3$. For $p = m$, Theorem 2.1 costs about 0.04 s per point.

Since $|\psi(\mathbf{t})|$ decays like $\|\mathbf{t}\|^{-\alpha}$, when $p = m - 1$ and n is close to $m - 1$ the excess $\alpha - p = (n - m + 1)/2$ over the integrability threshold is small, ψ is not smooth at $\mathbf{t} = \mathbf{0}_p$, and the inversion loses accuracy even with the extrapolation described above: for $m = 3$ and $p = 2$, for example, its relative error reaches 3×10^{-2} for $n = 2.2$ and 2×10^{-4} for $n = 4$, whereas the marginalization (76) and, for $p = 1$, the Bessel representation are unaffected. The programs that accompany the paper therefore use Theorem 2.1 for $p = m$; the marginalization (76) for $p = m - 1 \geq 3$ when $\mathbf{\Sigma}_z$ is not proportional to $\mathbf{\Sigma}_w^{-1}$, and for $p = m - 1$ when $n < m + 2$ (the Bessel representation when $m = 2$); and the characteristic function representation in the other cases with $p \leq 3$ or $\mathbf{\Sigma}_z \propto \mathbf{\Sigma}_w^{-1}$, together with the diagnostics of the inversion described at the beginning of this section. The remaining cases, $4 \leq p \leq m - 2$ with $\mathbf{\Sigma}_z$ not proportional to $\mathbf{\Sigma}_w^{-1}$, are handled by the conditional Monte Carlo estimator (80): for $p = m - 2$ the marginalization (76) is a four-dimensional integral, but it takes many minutes per point, and for $4 \leq p \leq m - 3$ every representation requires an integral of dimension at least five.

Table 3: Computing time (seconds) of the density of $\mathbf{q} = \mathbf{L}'\mathbf{W}\mathbf{z}$ at one point and on a grid of 101 points, by representation, for $n = m + 5$ and the general and identity settings of Table 2. CF: Corollary 4.1, in the reduced form of Remark 4.5 in the identity setting; Bessel: Sections 3.1–3.2; Marg.: marginalization (76) of Theorem 2.1; Prop.: Section 3.3; Thm 2.1: Theorem 2.1. Two numbers give the time at one point and on the grid; for the Bessel, marginalization, and Section 3.3 representations only the time at one point is reported, because they share no computation across evaluation points, so that a grid of 101 points takes about 101 times as long. A dash indicates that the representation is not applicable.

m	p	Setting	Thm 2.1	CF	Bessel	Marg.	Prop.
2	1	General	–	0.1 / 0.2	0.09	10	–
3	1	General	–	0.1 / 0.2	0.06	–	–
5	1	General	–	0.1 / 0.2	0.05	–	–
10	1	General	–	0.1 / 0.2	–	–	–
3	2	General	–	4.6 / 4.8	12	12	–
4	2	General	–	5.2 / 5.2	11	–	–
5	2	General	–	5.8 / 6.0	11	–	–
4	3	General	–	90 / 92	–	12	–
5	3	General	–	109 / 109	–	–	–
5	4	General	–	–	–	14	–
5	5	General	0.03 / 3.1	–	–	–	–
10	10	General	0.04 / 3.4	–	–	–	–
3	1	Identity	–	0.01 / 0.02	0.07	–	0.2
5	1	Identity	–	0.01 / 0.02	0.08	–	0.2
5	2	Identity	–	0.8 / 1.1	16	–	10
5	3	Identity	–	1.0 / 3.0	–	–	11
5	4	Identity	–	1.7 / 5.1	–	13	7.7
10	2	Identity	–	0.8 / 0.8	–	–	17
10	5	Identity	–	0.8 / 0.8	–	–	14
10	9	Identity	–	0.8 / 2.3	–	16	1.7

6. Summary

The Wishart and normal distributions are fundamental in multivariate statistical theory, where they arise in both parameter estimation and hypothesis testing (see Muirhead, 1982). These matrix-variate and multivariate families are well studied (see, e.g., Muirhead, 1982; Gupta and Nagar, 2000) and underlie many distributional families, such as elliptically contoured distributions (Gupta et al., 2013), skew-normal distributions (Azzalini, 2013), and the Laplace distribution (Kotz et al., 2012). Their properties are also widely used to derive theoretical results in applications such as finance (Bodnar et al., 2022; Kan et al., 2024) and biology (Serván et al., 2025).

In this paper, we derived the density function of the product of a Wishart matrix and a normal vector, expressed as a two-dimensional integral. The result is obtained in the general case, without assuming any relationship between the covariance matrices of the Wishart and normal distributions, and in particular without assuming proportionality. We also derived the density of linear combinations of the product, with particular attention to the special cases of one and two linear combinations and of $\mathbf{\Sigma}_z$ proportional to $\mathbf{\Sigma}_w^{-1}$; the resulting expressions are given as low-dimensional integrals that can be evaluated numerically. In addition, we showed that the characteristic function of any $p \leq m$

linear combinations is a one-dimensional integral and bounded its rate of decay, so that the density of $p < m$ linear combinations can be recovered by Fourier inversion as a $(p+1)$-dimensional integral; combined with the marginalization of the two-dimensional representation of the full product, the density of any p linear combinations is available from an integral of dimension at most $\min(p+1, m-p+2)$, and of dimension at most three when $\mathbf{\Sigma}_z$ is proportional to $\mathbf{\Sigma}_w^{-1}$. The representations were validated against simulation, and the numerical comparison of Section 5 indicates which representation is the fastest in each configuration.

Several extensions are left for future work, including the moments and tail behavior of the proposed matrix-variate mixture family, a singular-Wishart counterpart in the spirit of Srivastava (2003), Bodnar et al. (2015), and Bodnar et al. (2018), and applications to the sampling distributions of statistics involving products of Wishart matrices and normal vectors in financial econometrics.

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A. Auxiliary results

A.1. Proof of Lemma 2.1

The proof of Lemma 2.1 uses the following lemma and corollary.

Lemma A.1. *Let* $\mathbf{x} \sim N_m(\mathbf{0}_m, \mathbf{I}_m)$, $\mathbf{W} \sim W_m(n, \mathbf{I}_m)$, $m \geq 2$, $n > m-1$, *and let* $\mathbf{x}$ *and* $\mathbf{W}$ *be independent. Define* $\mathbf{h} = \mathbf{W}^{-1/2}\mathbf{x}$, $\mathbf{P} = [\tilde{\mathbf{h}},\ \mathbf{P}_1]$ *orthonormal matrix with* $\tilde{\mathbf{h}} = \mathbf{h}/\sqrt{\mathbf{h}'\mathbf{h}}$, *and*

$$\mathbf{H} = \mathbf{P}'\mathbf{W}\mathbf{P} = \begin{pmatrix} H_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{12}' & \mathbf{H}_{22} \end{pmatrix}, \quad H_{11} = \tilde{\mathbf{h}}'\mathbf{W}\tilde{\mathbf{h}}. \tag{83}$$

Let $v_{11} = H_{11}$, $\mathbf{v}_{12} = \mathbf{H}_{12}'/\sqrt{H_{11}}$, $\mathbf{V}_{22\cdot1} = \mathbf{H}_{22} - \mathbf{H}_{12}'\mathbf{H}_{12}/H_{11}$. *Then, it holds that*

(i) $(\mathbf{h}, v_{11})$, $\mathbf{v}_{12}$, $\mathbf{V}_{22\cdot1}$ *are mutually independent;*

(ii) $\mathbf{h}|v_{11} \sim N_m(\mathbf{0}_m, \frac{1}{v_{11}}\mathbf{I}_m)$;

(iii) $v_{11} \sim \chi^2_{n-m+1}$;

(iv) $\mathbf{v}_{12} \sim N_{m-1}(\mathbf{0}_{m-1}, \mathbf{I}_{m-1})$;

(v) $\mathbf{V}_{22\cdot1} \sim W_{m-1}(n, \mathbf{I}_{m-1})$.

Proof of Lemma A.1. Following the assumptions of the lemma, the joint density of $\mathbf{h} = \mathbf{W}^{-1/2}\mathbf{x}$ and $\mathbf{W}$ is given by

$$f(\mathbf{h}, \mathbf{W}) = f(\mathbf{h}|\mathbf{W})f(\mathbf{W}) = \frac{|\mathbf{W}|^{1/2}}{(2\pi)^{m/2}} \exp\left(-\frac{1}{2}\mathbf{h}'\mathbf{W}\mathbf{h}\right) \frac{|\mathbf{W}|^{(n-m-1)/2}}{2^{mn/2}\Gamma_m\left(\frac{n}{2}\right)} \exp\left(-\frac{1}{2}\mathrm{tr}(\mathbf{W})\right), \tag{84}$$

where $\Gamma_m(\cdot)$ denotes the multivariate gamma function (see Muirhead, 1982, Theorem 2.1.12).

Let $\mathbf{P} = [\tilde{\mathbf{h}},\ \mathbf{P}_1]$ be an orthonormal matrix with $\tilde{\mathbf{h}} = \mathbf{h}/\sqrt{\mathbf{h}'\mathbf{h}}$. Making transformation $(\mathbf{h}, \mathbf{W})$ to $(\mathbf{h}, \mathbf{H})$, $\mathbf{H} = \mathbf{P}'\mathbf{W}\mathbf{P}$, with Jacobian equal to one, we get

$$f(\mathbf{h}, \mathbf{H}) = \frac{1}{(2\pi)^{m/2}} \exp\left(-\frac{1}{2}(\mathbf{P}'\mathbf{h})'\mathbf{H}(\mathbf{P}'\mathbf{h})\right) \frac{1}{2^{mn/2}\Gamma_m\left(\frac{n}{2}\right)} |\mathbf{H}|^{(n-m)/2} \exp\left(-\frac{1}{2}\mathrm{tr}(\mathbf{H})\right). \tag{85}$$

Using the definitions of $v_{11}, \mathbf{v}_{12}, \mathbf{V}_{22\cdot1}$ and the fact that the Jacobian of the transformation from $(\mathbf{h}, H_{11}, \mathbf{H}_{12}, \mathbf{H}_{22})$ to $(\mathbf{h}, v_{11}, \mathbf{v}_{12}, \mathbf{V}_{22\cdot1})$ is $v_{11}^{(m-1)/2}$, we obtain

$$f(\mathbf{h}, v_{11}, \mathbf{v}_{12}, \mathbf{V}_{22\cdot1}) = \frac{v_{11}^{m/2}}{(2\pi)^{m/2}} \exp\left(-\frac{1}{2}v_{11}\mathbf{h}'\mathbf{h}\right) \times \frac{v_{11}^{(n-m+1)/2-1}}{2^{(n-m+1)/2}\Gamma\left(\frac{n-m+1}{2}\right)} \exp\left(-\frac{v_{11}}{2}\right)$$

$$\times \frac{|\mathbf{V}_{22\cdot 1}|^{(n-m)/2}}{2^{(m-1)n/2}\Gamma_{m-1}\left(\frac{n}{2}\right)} \exp\left(-\frac{1}{2}\mathrm{tr}(\mathbf{V}_{22\cdot 1})\right) \times \frac{1}{(2\pi)^{(m-1)/2}} \exp\left(-\frac{\mathbf{v}_{12}'\mathbf{v}_{12}}{2}\right), \tag{86}$$

where the properties of the multivariate gamma function is used (see page 95 in Muirhead, 1982). The formula of $f(\mathbf{h}, v_{11}, \mathbf{v}_{12}, \mathbf{V}_{22\cdot 1})$ shows that $(\mathbf{h}, v_{11})$, $\mathbf{v}_{12}$, $\mathbf{V}_{22\cdot 1}$ are mutually independent and have the distributions as summarized in the statement of the lemma. □

Corollary A.1. *Let* $\mathbf{x} \sim N_m(\mathbf{0}_m, \mathbf{I}_m)$, $\mathbf{W} \sim W_m(n, \mathbf{I}_m)$, $m \geq 2$, $n > m-1$, *and let* $\mathbf{x}$ *and* $\mathbf{W}$ *be independent. Define* $\mathbf{h} = \mathbf{W}^{-1/2}\mathbf{x}$, $\mathbf{P} = [\tilde{\mathbf{h}},\ \mathbf{P}_1]$ *orthonormal matrix with* $\tilde{\mathbf{h}} = \mathbf{h}/\sqrt{\mathbf{h}'\mathbf{h}}$, *and*

$$\mathbf{H} = \mathbf{P}'\mathbf{W}\mathbf{P} = \begin{pmatrix} H_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{12}' & \mathbf{H}_{22} \end{pmatrix}, \quad H_{11} = \tilde{\mathbf{h}}'\mathbf{W}\tilde{\mathbf{h}}. \tag{87}$$

Let $v_{11} = H_{11}$, $\mathbf{v}_{12} = \mathbf{H}_{12}'/\sqrt{H_{11}}$, $\mathbf{V}_{22\cdot 1} = \mathbf{H}_{22} - \mathbf{H}_{12}'\mathbf{H}_{12}/H_{11}$. *Then, we get the following stochastic representations*

$$\mathbf{h} \overset{d}{=} \frac{\mathbf{y}}{\sqrt{v_1}}, \tag{88}$$

$$|\mathbf{W}| \overset{d}{=} \prod_{i=1}^{m} v_i, \tag{89}$$

where $\mathbf{y}$, v_i, $i = 1, \ldots, m$ *are mutually independent with* $\mathbf{y} \sim N_m(\mathbf{0}_m, \mathbf{I}_m)$ *and* $v_i \sim \chi^2_{n-m+i}$, $i = 1, \ldots, m$.

Proof of Corollary A.1. The results of the corollary follow from Lemma A.1, the equalities $|\mathbf{W}| = |\mathbf{H}|$ and $|\mathbf{H}| = v_{11}|\mathbf{V}_{22\cdot 1}|$, and the fact that (see Theorem 3.2.15 in Muirhead, 1982)

$$|\mathbf{V}_{22\cdot 1}| \overset{d}{=} \prod_{i=2}^{m} v_i, \tag{90}$$

where v_i, $i = 2, \ldots, m$, are independent with $v_i \sim \chi^2_{n-m+i}$. □

Proof of Lemma 2.1. For brevity, we write $\mathbf{W}$ for $\tilde{\mathbf{W}}$. For $\mathbf{q} = \mathbf{0}_m$, we have $\boldsymbol{\xi} = \mathbf{0}_m$, and the lemma follows from (89). Let $\mathbf{q} \neq \mathbf{0}_m$, and let $\mathbf{Q} = [\tilde{\mathbf{q}},\ \mathbf{Q}_1]$ be an $m \times m$ orthonormal matrix whose first column is $\tilde{\mathbf{q}} = \mathbf{q}/(\mathbf{q}'\mathbf{q})^{\frac{1}{2}}$. Let

$$\mathbf{A} = \begin{bmatrix} A_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix} = (\mathbf{Q}'\mathbf{W}^{-1}\mathbf{Q})^{-1}, \tag{91}$$

with A_{11} univariate. Theorem 3.2.11 in Muirhead (1982) implies that $\mathbf{A} = (\mathbf{Q}'\mathbf{W}^{-1}\mathbf{Q})^{-1} \sim W_m(n, \mathbf{I}_m)$, and Theorem 3.2.10 of Muirhead (1982) yields

$$A_{11\cdot 2} = A_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21} \equiv v_1 \sim \chi^2_{n-m+1}, \tag{92}$$

$$-\mathbf{A}_{22}^{-\frac{1}{2}}\mathbf{A}_{21} \equiv \mathbf{x} \sim N_{m-1}(\mathbf{0}_{m-1}, \mathbf{I}_{m-1}), \tag{93}$$

$$\mathbf{A}_{22} \sim W_{m-1}(n, \mathbf{I}_{m-1}), \tag{94}$$

where v_1, $\mathbf{x}$ and $\mathbf{A}_{22}$ are mutually independent. In the derivation of (93), it is used that $\mathbf{A}_{21}|\mathbf{A}_{22} \sim N_{m-1}(\mathbf{0}_{m-1}, \mathbf{A}_{22})$ and, consequently, $\mathbf{x} = -\mathbf{A}_{22}^{-\frac{1}{2}}\mathbf{A}_{21}$ is unconditionally standard multivariate normally distributed, and it is independent of $\mathbf{A}_{22}$.

Hence, it follows that

$$|\mathbf{W}| = |\mathbf{A}| = A_{11\cdot 2}|\mathbf{A}_{22}| = v_1|\mathbf{A}_{22}|, \tag{95}$$

$$\tilde{\mathbf{q}}'\mathbf{W}^{-1}\mathbf{q} = (\mathbf{q}'\mathbf{q})^{\frac{1}{2}}\tilde{\mathbf{q}}'\mathbf{W}^{-1}\tilde{\mathbf{q}} = (\mathbf{q}'\mathbf{q})^{\frac{1}{2}}A_{11\cdot 2}^{-1} = \frac{(\mathbf{q}'\mathbf{q})^{\frac{1}{2}}}{v_1}, \tag{96}$$

$$\mathbf{Q}_1'\mathbf{W}^{-1}\mathbf{q} = (\mathbf{q}'\mathbf{q})^{\frac{1}{2}}\mathbf{Q}_1'\mathbf{W}^{-1}\tilde{\mathbf{q}} = -(\mathbf{q}'\mathbf{q})^{\frac{1}{2}}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}A_{11\cdot 2}^{-1} = \frac{(\mathbf{q}'\mathbf{q})^{\frac{1}{2}}\mathbf{A}_{22}^{-\frac{1}{2}}\mathbf{x}}{v_1}. \tag{97}$$

For $m > 2$, Corollary A.1, applied to $\mathbf{A}_{22} \sim W_{m-1}(n, \mathbf{I}_{m-1})$ and $\mathbf{x}$, gives, jointly, $|\mathbf{A}_{22}| \stackrel{d}{=} \prod_{i=2}^{m} v_i$ and $\mathbf{A}_{22}^{-\frac{1}{2}}\mathbf{x} \stackrel{d}{=} \mathbf{y}/\sqrt{v_2}$, where $\mathbf{y} \sim N_{m-1}(\mathbf{0}_{m-1}, \mathbf{I}_{m-1})$ and $v_i \sim \chi^2_{n-m+i}$, $i = 2, \ldots, m$, are mutually independent and independent of v_1. For $m = 2$, the same holds with $v_2 = A_{22} \sim \chi^2_n$ and $y = x$. Therefore, jointly, $|\mathbf{W}| \stackrel{d}{=} \prod_{i=1}^{m} v_i$ and

$$\mathbf{W}^{-1}\mathbf{q} = \mathbf{Q}\mathbf{Q}'\mathbf{W}^{-1}\mathbf{q} = \tilde{\mathbf{q}}\,(\tilde{\mathbf{q}}'\mathbf{W}^{-1}\mathbf{q}) + \mathbf{Q}_1(\mathbf{Q}_1'\mathbf{W}^{-1}\mathbf{q}) \stackrel{d}{=} \frac{1}{v_1}\left(\mathbf{q} + \frac{(\mathbf{q}'\mathbf{q})^{\frac{1}{2}}\mathbf{Q}_1\mathbf{y}}{\sqrt{v_2}}\right). \tag{98}$$

Since $\boldsymbol{\xi} = (\mathbf{q}'\mathbf{q})^{\frac{1}{2}}\mathbf{Q}_1\mathbf{y} \sim N_m(\mathbf{0}_m, (\mathbf{q}'\mathbf{q})\mathbf{Q}_1\mathbf{Q}_1')$ and $\mathbf{Q}_1\mathbf{Q}_1' = \mathbf{I}_m - \tilde{\mathbf{q}}\tilde{\mathbf{q}}'$, its covariance matrix is $(\mathbf{q}'\mathbf{q})\mathbf{I}_m - \mathbf{q}\mathbf{q}'$, which completes the proof. □

A.2. Density of a quadratic form in standard normal random variables

Suppose $\mathbf{x} \sim N_m(\mathbf{0}_m, \mathbf{I}_m)$ and let $Q = a_0 + \mathbf{a}_1'\mathbf{x} + \mathbf{x}'\mathbf{D}\mathbf{x}$, where $\mathbf{D} = \mathrm{Diag}(d_1, \ldots, d_m)$ and $d_1 \geq \cdots \geq d_m > 0$. With $\boldsymbol{\mu} = -\mathbf{D}^{-1}\mathbf{a}_1/2$, we have $Q = \tilde{Q} + a_0 - \boldsymbol{\mu}'\mathbf{D}\boldsymbol{\mu}$, where $\tilde{Q} = (\mathbf{x} - \boldsymbol{\mu})'\mathbf{D}(\mathbf{x} - \boldsymbol{\mu}) = \sum_{i=1}^{m} d_i X_i$ and $X_i \sim \chi^2_1(\delta_i)$, $i = 1, \ldots, m$, are independent noncentral chi-squared random variables with noncentrality parameters $\delta_i = a_{1i}^2/(4d_i^2)$. Hence $f_Q(q) = f_{\tilde{Q}}(q - a_0 + \boldsymbol{\mu}'\mathbf{D}\boldsymbol{\mu})$, and only the density of $\tilde{Q}$ is needed.

There are two approaches to computing $f_{\tilde{Q}}(\tilde{q})$. The first inverts the characteristic function of $\tilde{Q}$ (Imhof, 1961):

$$f_{\tilde{Q}}(\tilde{q}) = \frac{1}{\pi}\int_0^\infty \Re\left(e^{-\mathrm{i}t\tilde{q}}\psi_{\tilde{Q}}(t)\right)\mathrm{d}t, \qquad \psi_{\tilde{Q}}(t) = \prod_{j=1}^{m}(1 - 2\mathrm{i}td_j)^{-\frac{1}{2}}\exp\left(\frac{\mathrm{i}td_j\delta_j}{1 - 2\mathrm{i}td_j}\right), \tag{99}$$

where the principal branch of each square root is taken.

The second writes the density of $\tilde{Q}$ as a mixture of chi-squared densities. Ruben (1962) shows that for $0 < \beta < 2d_m$, we have

$$f_{\tilde{Q}}(\tilde{q}) = \frac{1}{\beta}\sum_{i=0}^{\infty} w_i f_{m+2i}\left(\frac{\tilde{q}}{\beta}\right), \tag{100}$$

where $f_v(x)$ is the density function of χ^2_v, and w_i can be recursively obtained using

$$w_0 = \exp\left(-\frac{1}{2}\sum_{j=1}^{m}\delta_j\right)\prod_{j=1}^{m}\left(\frac{\beta}{d_j}\right)^{\frac{1}{2}}, \tag{101}$$

$$w_i = \frac{1}{2i}\sum_{k=0}^{i-1} p_{i-k}\, w_k, \quad i \geq 1, \tag{102}$$

where, for $i \geq 1$,

$$p_i = \sum_{j=1}^{m}\left(1 - \frac{\beta}{d_j}\right)^i + i\sum_{j=1}^{m}\frac{\beta\delta_j}{d_j}\left(1 - \frac{\beta}{d_j}\right)^{i-1}. \tag{103}$$

For $0 < \beta < 2d_m$, the series converges and $\sum_{i=0}^{\infty} w_i = 1$; for $0 < \beta \leq d_m$, in addition, $w_i \geq 0$ for all i, so that $f_{\tilde{Q}}$ is a mixture of scaled chi-squared densities. The programs use $\beta = 0.90625\, d_m$.